Comprehensive Engineering Review: Permanent Magnet Manufacturing Waste Generation, Treatment, and Recovery

Summary

Sintered NdFeB magnet manufacturing exhibits unusually high internal material losses, with one study stating that NdFeB waste generated during manufacturing accounts for ~30 wt% of total production, and another reporting that up to 50% of raw materials may be wasted as scraps/swarf/sludge at magnet manufacturing sites. Multiple sources converge on machining-derived sludge/swarf as the dominant waste form, including reports that the "new solid scrap and cutting sludge (swarf)" accounts for 20–30% of the starting alloy and that a large quantity (~30%) of Nd_2Fe_14B sludge is produced in cutting and grinding.

From a resource and hazard standpoint, NdFeB machining sludges are consistently described as REE- and iron-rich: several sources state that these sludges contain ~20–30% REEs and ~60–70% Fe (dry basis), and grinding slurry REE content is reported up to 30 wt%. Facility-scale waste generation data from a U.S. DOE environmental assessment provides concrete magnitudes for a modern U.S. magnet-related facility: rare-earth sludge/grinding slurry at 16,600 kg/week (managed as hazardous waste) and spent coolant at 25 m^3/week (hazardous), both disposed at certified waste facilities.

On the liquid-effluent side, the dominant current-practice treatment method for metal-bearing wastewaters consistent with magnet coating (Ni-Cu-Ni electroplating) and machining is chemical precipitation (hydroxide and/or sulfide), using reagents including lime, NaOH, and sodium sulfide, typically with pH control in the ~9.0–9.5 window for hydroxide precipitation effectiveness for many metals. Downstream compliance drivers can be stringent: US EPA metal finishing categorical standards (40 CFR 433) specify total Cu at 3.38 mg/L daily max and 2.07 mg/L monthly average and total Ni at 3.98 mg/L daily max and 2.38 mg/L monthly average, while EU BAT-associated emission level (BAT-AEL) ranges can be much tighter for some parameters (e.g., Ni 0.01–0.2 mg/L, TSS 5–30 mg/L, COD 30–90 mg/L as excerpted).

Boron is a special-case contaminant for NdFeB facilities with Ni plating because Watts Ni baths commonly contain boric acid (30–45 g/L), and boron speciation (pKa ≈ 9.23) makes deep removal strongly pH-dependent and often poor at neutral pH in membrane systems; deep polishing frequently requires boron-selective resins (BSR) or high-pH multi-pass RO/NF configurations, with reported cases achieving sub-1 mg/L residual boron.

Methods

This report synthesizes quantitative engineering and regulatory data from the provided raw excerpts spanning (i) academic and technical literature on magnet manufacturing wastes and hydrometallurgical treatment of NdFeB machining sludges, (ii) government technical documentation (including a DOE environmental assessment for waste generation rates and waste handling practices), and (iii) regulatory/BAT documentation and vendor datasheets for wastewater-treatment and hydromet equipment. Only parameters explicitly present in the provided excerpts are reported as factual values; where a typical engineering calculation would normally be performed (e.g., complete plant-wide water balance or full WWTP equipment sizing), this report instead provides the governing design parameters and ranges available in the excerpts, and flags missing information as design-critical gaps.

Waste streams inventory

Waste streams in permanent magnet manufacturing arise in (A) powder handling and pressing, (B) sintering and heat treatment, (C) machining (cutting, grinding, lapping/polishing, EDM), (D) surface treatment and coating (including

Ni-Cu-Ni electroplating), and (E) quality rejection and ancillary materials. Across sources, machining is repeatedly identified as the dominant loss mechanism and the dominant generator of sludge/swarf waste, with 20–30% of starting alloy reported as "solid scrap and cutting sludge (swarf)" in one synthesis and ~30% sludge reported from cutting and grinding in other studies.

Powder handling and pressing

Powder-related waste includes oxidized/off-spec powder, green-body residues/press rejects, and airborne fines captured in dust collectors. A U.S. environmental assessment reports "oxidized powder and green body residues" at 500 kg/week stored in 55-gallon barrels and managed as non-hazardous in that table, disposed at a certified waste facility.

Powder size targets are in the few-micron regime. Jet milling is described as producing a narrow PSD around ~5 μm, and other sources describe jet-milled powder typically ~3–5 μm; one manufacturing guide describes powder particles required to be ~3–4 μm. Handling is commonly inerted; one process description states powder is kept in inert gas atmosphere and handled in glove boxes before automated pressing.

Powder reactivity is a primary safety and waste-handling driver: SDS statements note that dry powders of neodymium magnets oxidize, smolder, and burn rapidly in air/oxygen and recommend maintaining powders in water slurry or inert atmospheres (N_2 or Ar) to prevent spontaneous combustion; guidance for spills similarly recommends sweeping up dust and storing as slurry or in inerted sealed containers. In cobalt-containing systems (SmCo or Co-bearing NdFeB grades), respirable fine cobalt metal powder (≤10 μm) is described as "Acutely toxic by inhalation Category 1," which can influence classification and industrial hygiene controls for dust collector catch and housekeeping residues.

Airborne fines are typically captured by dust collectors; the same U.S. facility document describes dust collectors on metal powder handling equipment with permitted removal efficiency up to ~99% for PM10 and PM2.5 emissions, which provides a basis for estimating captured mass vs. residual stack emissions for permitting calculations.

Sintering and heat treatment

Sintering is described at 1000–1100 ℃, including a statement that the compact is sintered at 1000–1100 ℃ in vacuum to densify particles into the final sintered magnet. Annealing is described as two stages at 850–950 ℃ and 400–600 ℃ in one process overview. During sintering, linear shrinkage of about 7–25% is reported, which drives stock allowances for machining and fixture/boat design allowances to account for dimensional change.

Quality requirements contribute to internal reject and rework. A density requirement of >98% of theoretical density is explicitly stated as necessary in one reference, implying that density shortfall is a reject criterion post-sinter and/or post-anneal and can translate into scrap or reprocessing flows.

Reprocessing routes for solid scrap include comminution via hydrogen decrepitation; one source references comminution via hydrogen decrepitation and another states hydrogen decrepitation occurs at atmospheric pressure and room temperature in that context, providing a basis for low-temperature scrap embrittlement prior to milling and reuse.

Machining

Machining (cutting, grinding, EDM, lapping/polishing) dominates waste generation for sintered magnet geometries that require tight tolerances. A synthesis source states that "solid scrap and cutting sludge (swarf)" accounts for 20–30% of starting alloy, and other sources report ~30% $Nd_2Fe_{14}B$ sludge produced in cutting and grinding processes.

One review also states more than 70% of waste generated in manufacturing consists of sludge that is heavily oxidized and contaminated with carbon from lubrication, reinforcing that waste is often generated as high-solids slurry/sludge rather than dry chips.

SmCo machining can have high breakage losses; one source describes Sm–Co magnets as the most brittle RE-based magnets with machining failure rate up to 20%, linked to low fracture toughness (1.9–2.9 MPa·m^1/2).

Grinding sludge and coolant For NdFeB grinding, a water-based emulsion is described as "typically recommended." A detailed coolant example describes Mobil® Mobilcut 230 as mineral-oil and boric-acid based, water miscible, applied at 5% concentration, with throughput of 30–50 L/min per wheel; its concentrate composition is given as mineral oil balance, boric acid 10–20 wt%, ethanol 1–5 wt%, monoethanolamine oleate 1–5 wt%, rapeseed oil 1–5 wt%, and others <10 wt%. A separate coolant datasheet provides typical operating pH 9.2–10.0 and recommended usage concentration 5–15% in water, which affects WWTP alkalinity and pH equalization demands if coolant blowdown enters the central WWTP. Coolant guidance also notes that coolants may include biocides to prevent microbial growth that can compromise coolant effectiveness (a driver for coolant replacement and thus spent-coolant waste generation).

Representative grinding slurry solids composition (wt%) is explicitly reported in one table as Nd 19.5, Dy 2.4, Pr 4.9, B 0.8, Al 0.2, C 1.3, O 5.6, with Fe as balance. Another table reports post solid–liquid separation ranges including Nd 14.8–19.0 wt%, B 0.9–1.2 wt%, Dy 5.7–6.1 wt%, Pr 0.1–4.4 wt%, with moisture content 25.1–33.1% and volatile content in solids 0.7–4.1%. Across sources, grinding slurry REE content is reported up to 30 wt%, consistent with the multi-source description of ~20–30% REEs and ~60–70% Fe in machining sludges (dry basis).

Abrasive dilution can be material in sludge; one cited diamond abrasive particle size is 25–40 μm, which is relevant for dewatering/filter media selection and for dilution of metal-bearing solids with inert abrasive fines. A cited slurry-density reference reports 3150–4075 kg/m^3 (context suggests slurry or related solids densities), which can be used as a bounding value for pump sizing and sludge tank design (acknowledging density depends strongly on solids fraction and entrained air).

Generation-rate anchors include both facility and machine scale. The DOE environmental assessment reports rare-earth sludge/grinding slurry at 16,600 kg/week, classified hazardous and stored in 55-gallon barrels for disposal at a certified waste facility. A separate machining study reports slurry generation from one machine at 0.6–1.4 kg/h.

Reactivity hazards can persist in wet sludge. One SDS states aqueous abraded metal material / aqueous grinding sludge develops hydrogen, and warns that on drying out, pyrophoric and/or self-heating materials may be present; this drives conservative handling (keep wet, vent drums where needed, avoid heat accumulation) for dewatered cakes or sludge storage in totes/drums.

Oil-based grinding sludge can be dominated by organic fractions. One environmental statement reports grinding sludge contains 51% oil, and a separate sludge characterization reports 52.5% cooling reagent with 12.45% Nd, 34.35% Fe, and 0.48% B, illustrating that deoiling/organic removal can be a prerequisite to stable roasting/leaching and to accurate metal mass accounting.

Cutting swarf An XRF analysis reports RE-Fe-B grinding swarf (wt%) Fe 73.71, RE 25.51, Ni 0.57, Cu 0.21, providing a basis for estimating both value and impurity load into hydromet circuits (Ni/Cu as potential contaminants). A facility document reports "process materials (grinding & sawing)" at 1,100 kg/week stored in 55-gallon barrels and disposed at a certified waste facility (non-hazardous in that table). A machining safety note emphasizes that NdFeB swarf powder produced during machining needs liquid cooling or it may spontaneously combust, consistent with SDS statements for dry powder reactivity.

Wire EDM Wire EDM uses a brass wire electrode and a dielectric that both cools and flushes debris. One source describes wire EDM using a thin brass wire (0.010 in) as electrode for precise cuts with minimal kerf loss; another experiment reports CuZn37 brass wire of 0.25 mm diameter and 900 N/mm² tensile strength. EDM guidance states dielectric fluid is used to cool and flush debris, and another machining FAQ notes oil (cutting fluid) flowing through the center of the electrode to flush particles from the cut path, establishing that EDM generates a mixed waste (spent dielectric + metal fines + filter cake).

A kerf width of 250–270 µm is reported for a wire-EDM context, enabling direct conversion to material loss per cut length and, therefore, estimating the mass load captured on EDM filters for a known cutting plan and magnet density. A study reports a demagnetized outer layer ~35 µm depth under the highest EDM voltage/capacitance tested, which can affect scrap/rework if magnet performance is surface-sensitive and may influence post-EDM allowance decisions.

Lapping and polishing slurry Polishing/lapping can generate fine slurries. One magnetorheological "MCF" slurry composition is reported as 45 wt% water-based magnetic fluid, 40 wt% iron powder, 12 wt% abrasive grains, and 3 wt% α-cellulose, with example particle sizes including 10 nm magnetic fluid mean diameter, 7 µm carbonyl iron powder, and 1 µm Al_2O_3 abrasive grains. Operationally, one report states the MCF slurry lost 0.724 g after 60 min polishing and recommends adding 0.1 mL water or MF every 10 min, illustrating evaporation/drag-out losses and the need for makeup addition in open polishing systems.

Surface treatment and coating

Sintered NdFeB magnets typically require corrosion-protection coatings; one manufacturing description identifies the baseline as Ni-Cu-Ni electroplating. Upstream preparation includes deoiling and oxide removal. A patent describes hot dipped deoiling solution at 40 g/L for cleaning oil impurities from matrix surfaces, and also describes nitric acid cleaning with 1–10% mass fraction to remove rust and oxides. A separate ICP-OES analysis of obtained solutions reports very high dissolved metal concentrations (Fe 40–60 g/L and Nd 10–20 g/L), indicating that certain acidic waste streams (e.g., pickling or intentional scrap dissolution) can be extremely concentrated and should be segregated and metered to prevent WWTP shock loading.

Rinsing is a major wastewater volume driver in plating and surface treatment. The STM BREF reference value for water usage is 3–20 L/m² of substrate surface per rinse stage, and guidance recommends a minimum of three stages of cascade rinsing with air agitation, which collectively can be used to estimate rinse water demand per plated area and to justify counter-current/cascade designs for drag-out minimization.

Ni-Cu-Ni electroplating chemistry The major compliance problem for nickel platers is controlling Ni and other metallic ions in effluents; drag-out drives rinse-water contamination and treatment requirement. Typical Watts nickel bath composition and operating conditions are reported as: NiSO_4·6H_2O 240–300 g/L, NiCl_2·6H_2O 30–90 g/L, H_3BO_3 30–45 g/L, temperature 40–60 ℃, pH 3.5–4.5, cathode current density 2–7 A/dm², deposition rate 25–85 µm/h. Overlapping ranges are also reported (e.g., NiSO_4 225–400 g/L, NiCl_2 30–60 g/L, H_3BO_3 30–45 g/L).

A copper bath variant described in a patent uses cupric pyrophosphate 20–120 g/L, potassium pyrophosphate 100–300 g/L, additives 0.1–50 g/L, with pH 7.0–10.0 in the described system.

Bath maintenance can generate carbon and anode sludges. Activated carbon treatment is described at 4–8 g/L with temperature around 66 ℃, agitation for 1 hour, and settling for 8–16 hours, implying batch side-tank requirements and sludge generation proportional to bath volume and treatment frequency.

Quality rejection and ancillary wastes

A facility document reports production byproduct (off-spec magnets) as 45 barrels (55-gallon), recycled within the plant, indicating an internal loop for solid scrap/rework rather than off-site disposal in that case. The same DOE environmental assessment provides a facility-scale combined sanitary sewer discharge of 7,905 gallons/day, including 4,755 gallons/day of waste process water and 3,150 gallons/day of domestic waste, which provides a concrete order-of-magnitude for daily wastewater flows (noting it is reported as sanitary sewer discharge in that document). For solvent-containing operations, the same facility document reports residual solvent waste of 240 gallons/week, managed as hazardous and disposed/recycled via certified facilities.

Treatment processes

Wastewater treatment

The dominant referenced approach for metal-bearing wastewater is chemical precipitation. EU BAT conclusions for water emissions note chemical precipitation and list typical chemicals used for metal precipitation as lime, sodium hydroxide, and sodium sulphide. A practical precipitation mechanism description states that adding caustic (NaOH) to water containing dissolved metals forms metal hydroxide solids, and that for most metals this reaction typically occurs when pH is in the range 9.0–9.5, which provides a control setpoint basis for mixed-metal precipitation systems treating Ni/Cu/Fe-bearing rinse waters and machining washwaters. Nickel-plating guidance similarly states the most widely used method of removing metallics from effluents is to precipitate nickel and other metals in sludges as hydroxides or sulfides.

For nickel-bearing rinse waters, the Nickel Institute handbook describes conventional treatment as pH adjustment to 9.3–10.0 to precipitate nickel as nickel hydroxide, followed by flocculation, clarification, and production of a wet sludge—i.e., a standard rapid-mix/neutralization stage followed by floc basins and a clarifier or lamella settler.

Where polishing or recovery is needed, BREF guidance mentions ion exchange and provides representative resin capacities (strong acid cation 1800 meq/L and chelating resin 1000 meq/L, sodium form). The Nickel Institute handbook also references evaporation, reverse osmosis, and electrodialysis as concentration/recovery techniques for rinse water, which can be used for rinse-water minimization, metal recovery, or to reduce discharge load.

Sludge dewatering and solids handling

Metal precipitation generates hydroxide sludges requiring dewatering. One filter press guidance source reports that achievable cake solids for metal hydroxide sludges can range from ~25% up to 50–65% depending on the metal hydroxide, and a filter press reference table lists "metal hydroxide sludge" typically achieving 30–55% cake solids. One guidance document states filter cake presses should be operated at not less than 7 bar and preferably 10–15 bar to reduce mass/volume/water content; another source distinguishes low pressure (100 psi ≈ 7 bar) and high pressure (225 psi ≈ 16 bar). Membrane diaphragm technology is reported as producing cakes up to 75% solids in half the cycle time of conventional presses, representing a dewatering-performance upgrade option for sludge logistics and disposal volume reduction.

Plating sludge and filter cake are described as hydrated hydroxide of iron (from work) and plating metals, together with soil, oil, and greases from preparation processes, which implies that upstream oil control (DAF/emulsion splitting) can meaningfully influence sludge properties and disposal classification. A WWTP sludge composition table for a plant example reports (g/kg dry): Zn 198, Pb 0.4, Cd 0.1, As 0.1, Fe 129, CaO 40, illustrating that sludge disposal cost and hazard profile are often driven by metals content and leachability rather than bulk mass alone.

Coolant management

Spent coolant (metalworking fluid) is often managed as hazardous waste. A facility document reports spent coolant at 25 m^3/week stored in a tank and disposed at a certified waste facility (hazardous). Operational monitoring triggers include tramp oil control and microbiological contamination: one guide recommends tramp oil should not exceed 2% contamination, and that immediate actions are required if bacterial contamination exceeds 10^6 cfu/mL, potentially requiring disposal of the MWF and full system clean; contaminated cutting fluid microbial concentrations are reported across 10^4 to 10^10 CFU/mL, which provides a quantitative basis for coolant-life and disposal frequency drivers beyond chemistry drift alone.

Spent MWF treatment approaches include chemical emulsion splitting (a chemical agent splits emulsion into oil and water phases) and thermal volume reduction (evaporating vessels drive off water, leaving reduced volume for hazardous disposal), which can be integrated upstream of WWTP to reduce hydraulic load and separate oily residues for off-site management.

Boron polishing

Boron removal is strongly pH-dependent due to boron speciation. A cited value of pKa = 9.23 indicates that speciation shifts near pH ~9.2; a review explicitly states RO can completely remove boron under alkaline conditions but only ~50% under neutral/acidic conditions because non-ionized boric acid dominates under those conditions. Boron-selective resin methods are reported at 93–98% removal, and BSR is described as capable of >99% removal, supporting BSR as a final polishing barrier for stringent discharge cases (e.g., 1 mg/L).

Quantitative membrane anchors for deep boron removal include two-pass RO with pH adjustment: SW-30 membrane flux 50 L·m^-2·h^-1 achieving 99% rejection at feed boron 50 mg/L and pH 10; staged NF can reduce boron from 3.87 to 0.84 mg/L at pH 10.5, and a two-stage NF with sorbitol complexation at pH 9.8 reportedly achieved <0.1 mg/L final effluent from 2.174 mg/L influent.

For boron-selective resins, a column adsorption study reports effective boron removal at space velocity 4 h^-1 and that adsorbed boron could be effectively eluted using 2 M HCl; eluate boron concentrations reported for different N-methylglucamine resins reach ~0.9–2.2 g/L depending on resin and brine case, which is directly relevant to regenerant brine management design.

Costs and economics

Disposal and treatment cost anchors

The provided dataset contains several cost anchors relevant to engineering-level screening but not sufficient for a complete facility TEA without site-specific waste characterization and vendor quotes. A general industrial wastewater CAPEX reference indicates that depending on volume and complexity, industrial wastewater treatment costs could be as low as $400,000 but easily exceed $1.5 million, and complex multi-stage streams can surpass a million dollars; this provides an order-of-magnitude CAPEX envelope for WWTP scope selection (equalization, precipitation, dewatering, polishing).

For plating-related waste management, the STM BREF provides a specific cost summary for a plating plant example: personnel costs ~EUR 138,000/year, energy costs ~EUR 53,000/year (plating shop), capital outlays EUR 300,000 for the wastewater treatment plant, and recycling costs including transport of EUR 165/t.

Generic hazardous waste disposal pricing is also provided for context: used oil disposal is cited as $0–$5 per gallon, while a generic "expensive" category is cited as $15–$50+ per gallon and a "moderate cost" category as $5–$15 per

gallon in one cost blog reference; these ranges are not magnet-specific but are useful as screening values for spent fluids and solvent wastes that are commonly shipped off-site.

Value concentration in machining wastes

NdFeB machining sludges concentrate REEs into a smaller mass, and multiple sources state these sludges contain ~20–30% REEs and ~60–70% Fe (dry basis). This concentration, combined with high waste generation fractions (20–30% of starting alloy as swarf/sludge), explains why recovery is frequently evaluated as both a disposal-cost-avoidance measure and a potential revenue stream from recovered REO products.

REE recovery process cost anchors

A techno-economic estimate for an NdFeB swarf-to-REO process reports total cost estimated at $1,092,585–$1,525,596/year, equivalent to ~$11 per kg of swarf processed or ~$29–$36 per kg mixed REO produced, with the process scaled to model processing capacity of 100 metric tons of NdFeB magnet swarf per year producing 32 metric tons of REOs per year. For broader REE projects, a separate TEA excerpt reports total CAPEX $126 million, total OPEX $122/kg, annual REE production 825 t/yr, REE revenue $178/kg, and gross profit $56/kg (context: a specific modeled REE production scenario).

Reagent and consumables are explicitly described as the largest operating expenses for mining projects (followed by power, labor, mining, G&A, and maintenance), which is directly applicable to swarf-to-REO hydromet circuits that consume acid, base, oxidants, and oxalic acid at scale. Unit price anchors in the provided sources include hydrochloric acid at approximately USD 137/MT (quarterly average), hydrogen peroxide at 816 USD/metric ton (June 2025 US), oxalic acid listings at US$500–700 (listing) and broader ranges reported elsewhere, and D2EHPA quoted at $4.0–5.0 (FOB, as presented).

Product price anchors include Nd_2O_3 projected at $56,500/metric ton (2024) and $77,500/metric ton (2025) and dysprosium oxide at USD/kg 340.00 in one PEA excerpt and 263.34 USD/kg in Jan 2026 Northeast Asia in another pricing report excerpt (indicating substantial variability by date and regional basis).

Regulatory requirements

Wastewater discharge limits

US EPA categorical metal-finishing effluent limits (40 CFR 433) in the provided excerpt include total copper 3.38 mg/L (daily max) and 2.07 mg/L (monthly average) and total nickel 3.98 mg/L (daily max) and 2.38 mg/L (monthly average). These values are directly relevant to NdFeB plants with Ni-Cu-Ni coating operations because bath drag-out contaminates rinse waters with Ni and Cu salts and boric acid (H_3BO_3).

EU BAT conclusions excerpts show much lower numeric ranges for some parameters and metals, including Ni 0.01–0.2 mg/L, Zn 0.05–1 mg/L, Fe 1–5 mg/L, TSS 5–30 mg/L, and COD 30–90 mg/L in the cited table excerpts, implying that a design targeting EU direct-discharge BAT-AEL performance may require more aggressive precipitation, polishing filtration, and potentially ion exchange or membrane polishing depending on permit-specific requirements.

China and Japan numeric limits are included in the provided excerpts for several parameters. For example, a China table excerpt lists total copper 0.5 mg/L (and other columns) and total zinc 2 mg/L in one excerpted table, while Japan national effluent standards excerpts include copper 3 mg/L and zinc 2 mg/L, dissolved iron 10 mg/L, and boron 10 mg B/L for non-coastal areas (and 230 mg B/L for coastal areas).

Boron as a rate-limiting contaminant

A review excerpt notes "stringent discharge limits set at 1 mg/L" for boron in certain contexts, and multiple boron-treatment sources demonstrate sub-1 mg/L performance under specific conditions; this combination makes boron a plausible rate-limiting contaminant when Ni plating is present and when discharge permits require ~1 mg/L boron, because conventional metals precipitation targets Ni/Cu/Fe but does not inherently guarantee deep boron removal without a boron-selective barrier or high-pH membrane strategy.

Metal mass balance

The provided excerpts do not include a complete plant-wide metal mass balance (inputs vs. outputs for Nd, Pr, Dy, Tb, Fe, B) for a specified facility capacity; however, the excerpts are sufficient to build a defensible first-order mass balance on a "starting alloy basis" and a "machining waste basis."

First-order alloy-to-waste allocation

One source states that solid scrap and cutting sludge (swarf) accounts for 20–30% of the starting alloy. Separately, multiple sources state machining sludges contain ~20–30% REEs and ~60–70% Fe (dry basis). Combining these two anchored statements yields a simple engineering estimate for REE losses to machining wastes per tonne of starting alloy:

- If 1.0 t starting alloy generates 0.20–0.30 t swarf/sludge (dry basis approximation), then REE in swarf/sludge = 0.20–0.30 t × (0.20–0.30) = 0.04–0.09 t REEs (40–90 kg REEs) per tonne starting alloy, by direct multiplication of the cited ranges.

This calculation is a first-pass sizing input for recovery systems and for valuing internal losses, but it requires site-specific correction for (i) moisture and coolant fraction (which can exceed 50% of as-generated sludge mass in some datasets), (ii) degree of oxidation, and (iii) whether "starting alloy" is defined on a pre-sinter or post-sinter basis for the specific factory routing.

Composition vectors for element tracking

Several explicit sludge/waste compositions are available for building element-level balances. Examples include a magnet sludge composition of 61.2% Fe, 24.1% Nd, 4.5% Dy, 1.0% B (mass%); a sintered waste composition of 59.81% Fe, 23.97% Nd, 7.124% Pr, 0.261% B, 0.062% Dy; and a separate waste composition including Fe 66.27%, Nd 23.9%, Pr 7.38%, Dy 0.662%, B 0.877%, Co 0.773%. These vectors can be used as feed assays for recovery-circuit mass balance closure and for estimating product split (NdPr oxide vs Dy oxide) as a function of grade and waste stream type.

Gaps and recommendations

Data gaps that block treatment system design

The provided excerpts contain strong point values for several unit operations and waste streams, but key design data for a bankable WWTP or integrated recovery system are missing. In particular, the excerpts do not provide (i) combined wastewater characterization (TSS, total metals by element, COD, oil and grease) for a defined magnet plant boundary, (ii) TCLP/leachability results for magnet sludge or WWTP sludge in a given jurisdiction, (iii) full hydraulic profiles for each segregated waste stream (machining blowdown vs plating rinse vs pickling dumps), and (iv) site-specific sludge dewatering performance (polymer dose, cake solids, filtrate quality) for NdFeB sludge matrices rather than generic hydroxide sludge ranges.

Recommendations for near-term engineering actions

A practical engineering sequence supported by the excerpts is to segregate high-value and high-reactivity solids from the WWTP, treat liquids via precipitation/clarification/dewatering, then add boron polishing and (optionally) a parallel sludge-to-REO recovery line. This architecture is supported by the unusually high REE content in machining sludges (20–30 wt% REEs) and by the demonstrated role of oxidative roasting in suppressing Fe dissolution and enabling selective leaching and high REE purity/yields in recovery flowsheets.

For boron compliance at ~1 mg/L, the excerpts support selecting a boron-selective resin or high-pH multi-pass RO/NF barrier as a polishing step, with resin regeneration designed around documented eluate concentrations (0.9–2.2 g/L B) and documented elution chemistries (e.g., 2 M HCl).

For REE recovery development, the excerpts support a staged pilot program structure (24 weeks total) with explicit decision gates of proceeding to FEED if recovery >90% and product >99%, and budget ranges of $500–700k (low) to $800k–$1.2M (high), with mini-plant throughput 25–50 kg/day dry sludge and a steady-state run target of 500 hours for mass balance closure.

Vendor and equipment notes

Where packaged equipment is required, vendor datasheets in the excerpts provide several concrete specification anchors. For filter presses, ANDRITZ reports filtration pressures 7–30 bar and plate sizes up to 2,500×2,500 mm with filtration areas up to 2,000 m^2; Diefenbach positions filter presses for highly acidic/corrosive sludges; and a general wastewater filter press source lists metal hydroxide sludge cake solids 30–55%.

For oil removal and suspended solids control upstream of precipitation, DAF vendor data include Evoqua RT-series rentals handling 100–1000 gpm and a nominal 200 gpm/400 gpm max effluent water rating in one Xylem rental listing, with stainless construction options; these values can be used to screen package-unit sizing for machining coolant blowdown or for combined oily wastewater streams prior to equalization and precipitation.

For chemical dosing and automation (critical to pH control in precipitation and to consistent polymer dosing for dewatering), Watson-Marlow Qdos pumps are described as designed for dosing lime, NaOH, $FeCl_3$, and polymers, with series flow rates up to 600 L/hr in one excerpt, supporting skid-based chemical feed systems for smaller WWTPs and pilot plants.

Process alternatives

The excerpts include both hydromet and high-temperature alternatives for REE recovery. Hydromet routes include full dissolution (e.g., 4 M HCl dissolution of demagnetized magnet particles for 24 h at room temperature) and selective leaching after oxidative roasting (e.g., oxidative roasting at 900 ℃ for 6 h followed by selective leaching achieving >99% REE and <5% Fe recovery in one report). High-temperature alternatives include molten chloride salt extraction ($MgCl_2$–KCl) at 700–1000 ℃ with ~90% REE recovery yield, and liquid Mg extraction at ~1000 ℃ described as slow (24–72 h).

For boron, alternatives include electrocoagulation (EC) with performance dependent on current density and contact time and energy/operating cost penalties at higher current densities, and hybrid adsorption–membrane systems demonstrating reduction to ~0.43 mg/L boron in one case (below a cited 1.0 mg/L discharge limit context).

Limitations

This report is constrained to the numeric excerpts provided and therefore cannot present a fully specified WWTP design (e.g., tank volumes, surface loading rates, chemical dose curves, sludge mass flow closure) for a specific magnet factory without additional influent and waste-stream characterization data. The report also cannot assign waste codes (EWC/RCRA) or definitive hazardous/non-hazardous classifications beyond those explicitly stated in the facility document excerpts, because such classifications generally require site-specific analytical results and jurisdiction-specific determinations.

Boron Polishing Technologies for NdFeB Magnet Manufacturing Wastewater to <1 mg/L B

1. Executive summary

Industrial boron control to sub-mg/L levels is routinely framed around a <1.0 mg/L discharge/quality target, with some standards cited at 0.5–1.0 mg/L depending on jurisdiction and end use. Multiple technology families have demonstrated the ability to reach <1 mg/L boron in treated water under specific conditions, including boron-selective sorbents/resins (reported as capable of >99% removal), multi-pass reverse osmosis with alkalinity management (reported 99% rejection at pH 10), and hybrid adsorption–membrane or resin–membrane trains (reported permeate 0.43 mg/L and/or 96–99% removal).

The primary engineering implication for a magnet plant WWTP is that conventional clarification/metal hydroxide precipitation alone is typically insufficient for deep boron removal under neutral conditions; performance for RO is explicitly described as ~50% when non-ionized boric acid dominates, whereas alkaline conditions and/or complexation can drive rejection toward ~99% in referenced systems. As a result, a two-tier design is common in practice: (1) upstream solids/oil/metals control to protect polishing unit operations, followed by (2) a dedicated boron polishing block using either boron-selective media (ion exchange / chelating sorbents), high-pH (often 2-pass) RO/NF, or electro-driven membrane processes (EDI/BMED) depending on water reuse vs. discharge objectives and brine disposal constraints.

2. Boron chemistry

Boron removal is strongly pH-dependent in membrane systems because the neutral species dominates under neutral or acidic conditions, which is explicitly linked to poor RO performance (~50% removal) in the cited review. The same review reports that under controlled alkaline conditions (feed boron 50 mg/L, pH 10.0) and at a permeation flux of 50 $L \cdot m^{-2} \cdot h^{-1}$, an SW-30 RO membrane achieved ~99% boron rejection when operated in a boron–D-mannitol complex system (alcohol/boron molar ratio = 2). pH-driven speciation effects are also reported for forward osmosis (FO): increasing pH from 4 to 10 increased boron removal from 12%⟶90% (FTS H_2O membrane), 8%⟶84% (PSU), and 8%⟶76% (Aquaporin Inside).

From an engineering perspective, this establishes two practical levers for deep boron treatment in industrial wastewater polishing: (a) shift boron into a more rejectable/adsorbable form via pH management and/or complexation, and (b) apply a unit operation with intrinsic selectivity for boron (e.g., N-methylglucamine functional media or boron-selective resin-filled EDI).

3. Technology comparison

The table below summarizes representative performance and key operating numbers extracted from the provided sources, emphasizing the ability to meet <1 mg/L boron and the operating "knobs" most relevant to specifying a polishing system.

Technology	Demonstrated performance	Key operating conditions reported	Engineering notes for <1 mg/L B targets
Boron-selective resin and related sorbents	Resin methods reported at 93–98% removal; BSR described as capable of >99% removal	Column operation examples include space velocity 4 h^{-1} with elution by 2 M HCl	Strong candidate for "final polishing" to sub-mg/L when upstream metals/TSS are controlled; regenerant handling becomes the dominant waste stream consideration (acid/base and high-B regenerant).
High-pH RO and multi-pass RO	RO at alkaline conditions can reach ~99% rejection at pH 10 in the cited complexation case (SW-30, flux 50 $L \cdot m^{-2} \cdot h^{-1}$, feed B = 50 mg/L)	Neutral/acid conditions linked to ~50% removal due to non-ionized boric acid dominance	Effective when brine management is acceptable; pH/complexant dosing and downstream neutralization are key integration issues.
Nanofiltration hybrids	Double-stage NF: second stage influent pH 10.5 reduced boron 3.87⟶0.84 mg/L; two-stage NF with sorbitol complexation and pH 9.8 achieved final effluent <0.1 mg/L (from influent 2.174 mg/L)	pH elevation and complexation are explicit control parameters	Potentially lower pressure than RO with strong polishing when staged and chemically conditioned; concentrate still requires disposition.
Electrocoagulation	Example optimum performance: 98% at pH 7, current density 12.5 $mA \cdot cm^{-2}$, contact time 90 min	Other EC configurations reported: EC-Al 70% removal of 100 mg/L B at pH 8, 10 mm electrode distance, 60 min, 5.5 $mA \cdot cm^{-2}$; interelectrode distance 2⟶0.5 cm increased removal 73%⟶92% after 150 min	Attractive as pretreatment to reduce boron load and co-remove some metals/TSS; electrical cost sensitivity is reported (2.7× cost increase when CD increased 2.5⟶6.25 $mA \cdot cm^{-2}$ in one case)
Chemical precipitation and coagulation	96% removal using $Ca(OH)_2$ within 25 min; 95% at pH 8.12 using $Ca(OH)_2$ plus $Al_2(SO_4)_3 \cdot 18H_2O$	High removals are reported under the cited conditions, but performance is chemistry-specific (reagent selection and pH control are critical)	Potentially high sludge generation and chemical consumption; best suited for high-B, smaller-volume streams or as a "bulk reduction" stage before polishing.

Technology	Demonstrated performance	Key operating conditions reported	Engineering notes for <1 mg/L B targets
Adsorption on $Al(OH)_3$ and other media	$Al(OH)_3$ sorption reported 94.7 mg B/g at pH 9.0 (and 27.8 mg B/g at pH 10.5) in high-B wastewater conditions (980 mg/L at pH 9.0; 635 mg/L at pH 10.5)	Desorption feasible at pH<5.0 enabling reuse; ~55% boron recovery reported; net profit estimated 2.85 $/m^3 (case-specific)	Best aligned with high-boron side streams (e.g., segregated rinses) rather than low-mg/L polishing; recovery economics depend on site-specific chemical costs and boron value.
Hybrid adsorption–membrane	Adsorption (nano-magnetite) achieved 74.39% removal leaving 2.2 mg/L; subsequent membrane polishing achieved 0.43 mg/L	Adsorption contact time 250 min; membrane test included scouring aeration 5 L/min; modified (1.0 wt% TiO_2) membrane reported 94.87% removal to 0.42 mg/L	Demonstrates an integrated approach to reliably cross below 1 mg/L where adsorption alone does not; fouling control and solids management are design-critical.

4. Boron selective resins

Boron-selective ion exchange and chelating media are widely reported as high-performance options, with resin methods summarized as 93–98% removal in one review and boron-selective resin (BSR) described as capable of >99% removal. Amberlite IRA-743 is specifically referenced with ~85.1% boron elimination at pH 8 and an adsorption capacity of 6.27 mg/g in the cited comparison review. Separate work on brine screening reports boron-selective media with capacities "above 5 mg/g," with effective column removal at a space velocity of 4 h^{-1} and elution using 2 M HCl.

A critical design reality in discharge polishing is that resin performance and cycle time depend strongly on the inlet boron concentration. In a pilot test using AmberLite PWA-10 (N-methylglucamine functional group), a short operation cycle of 0.45 h was reported at very high inlet boron (121,000 ppm), whereas a longer cycle of 41.35 h was reported at lower inlet boron (3,800 ppm), with an average boron removal efficiency of 87.08% yielding 0.491 ppm in the reported case. For magnet manufacturing, this supports the standard engineering approach of source segregation and equalization (minimize "slug loads" of boron into the polishers) rather than relying on resin capacity alone to buffer extreme concentration variability.

Mechanism and pH window

The cited resin products are explicitly tied to N-methylglucamine functionality (e.g., AmberLite PWA-10) for boron selectivity. Reported performance for Amberlite IRA-743 is shown at pH 8 (~85.1% removal), and a separate batch study reports maximum performance at pH 9.5 (initial B 40 mg/L; resin dosage 110 g/L), consistent with strong pH sensitivity in practical operation.

Column process configuration

For polishing to <1 mg/L, lead–lag columns (or duty/standby columns) are typically used in practice, and the provided literature includes multiple column-operating-rate anchors that are usable in preliminary sizing. One column example for a sorbent system uses 2 bed volumes per hour (2 BV/h) at constant flow in a 15 mm i.d. glass column with 150 mm bed height, with regeneration reported using 0.32 mol/dm^3 NaOH (98.5% boron elution) and HCl as a secondary eluent for other ions in that system. Another resin column example for boron removal from brine is reported at 4 h^{-1} space velocity with 2 M HCl elution.

A separate mine-drainage case reports a salt-form boron-specific resin capacity of 1.5 mg B/mL resin at 10% boron leakage, at a reported flow of 161 h^{-1} (interpretable as very high space velocity), and discusses a strategy of reusing 90% of regenerant with associated boric acid production (4,400 tons/year) in that application context. While that specific tonnage is not directly transferable to a magnet facility, it is a clear indication that regenerant reuse and boron recovery can dominate resin OPEX and waste generation strategy when inlet concentrations are high and/or discharge limits are stringent.

Regeneration chemistry and secondary waste

Regeneration/elution is explicitly reported using strong acid and/or base in multiple sources, including 2 M HCl for elution in column screening and 0.32 mol/dm^3 NaOH for boron desorption from a packed bed (98.5% elution of boron). Another resin study reports regeneration by washing with 7% HCl and 4% NaOH for a boron sorbent (reported saturated adsorption capacity 5 mg/g in that referenced example). These data imply the key mass-balance consequence of resin polishing: boron is transferred from a dilute continuous wastewater stream into a smaller-volume, high-boron regenerant waste (acidic and/or caustic), which then becomes the main discharge/disposal design problem for the polishing block.

5. High pH reverse osmosis

The cited review explicitly states that RO can "completely remove the boron under alkaline conditions," but also that at neutral/acidic conditions the removal rate is "only about 50%" due to dominance of non-ionized boric acid molecules. This statement is directly aligned with practical membrane system design: if RO is used as the boron compliance lever, it generally requires either pH elevation and/or complexation chemistry to shift boron into a form with higher rejection.

Quantitatively, an SW-30 membrane is reported at ~99% rejection with feed boron 50 mg/L at pH 10 and flux 50 L·m^{-2}·h^{-1} in a boron–D-mannitol complex system (alcohol/boron molar ratio 2). The same review also reports that modified RO membranes can achieve boron rejection "more than 90%" (attributed to changes in pore size and surface properties in that cited context). In addition, a two-pass RO with pH adjustment is reported to deliver 99% rejection at 50 mg/L feed boron and pH 10 at flux 50 L·m^{-2}·h^{-1} (SW-30 membrane).

From an engineering tradeoff standpoint, RO performance is inherently linked to pressure/energy and concentrate handling, and one source notes a high pressure range of 50–75 bar to overcome osmotic pressure of ~26–29 bar in the cited RO context. For a magnet wastewater plant evaluating RO for boron compliance, the dominant decision variables therefore become (i) whether concentrate disposal is feasible, and (ii) whether upstream conditioning can reliably hold pH/chemistry in the needed window without destabilizing metals removal or causing scaling in the membrane train.

6. Electrocoagulation

Electrocoagulation (EC) is reported with wide performance ranges and strong dependence on current density, electrode configuration, and time. One reported "optimal performance" is 98% removal at pH 7, current density 12.5 mA/cm², and contact time 90 minutes. Another reported case for EC-Al is 70% removal from 100 mg/L boron at pH 8 with electrode distance 10 mm, time 60 min, and current density 5.5 mA/cm².

Hardware geometry and electrical setpoints have direct leverage on removal: decreasing interelectrode distance from 2 cm to 0.5 cm increased removal from 73% to 92% after 150 minutes in a reported case. Cost/energy sensitivity is also explicitly highlighted: increasing current density from 2.5 to 6.25 mA/cm² did not improve removal significantly in the cited case but increased cost by 2.7×.

For an engineer designing a <1 mg/L polishing block, the cited EC data suggest EC is best positioned as a pretreatment or load-reduction step (especially for higher-B side streams) rather than the sole compliance unit, unless the EC system is paired with a downstream polishing technology to guarantee consistent sub-mg/L residual boron under varying influent quality.

7. Magnesium and calcium precipitation

Precipitation/coagulation pathways are reported as capable of high boron removal under certain reagent and pH conditions. A cited example reports 96% boron removal using chemical precipitation with calcium hydroxide within 25 minutes. Another reports 95% removal at pH 8.12 using chemical precipitation and coagulation with $Ca(OH)_2$ and $Al_2(SO_4)_3 \cdot 18H_2O$.

For magnesium oxide, a mine-drainage study reports that MgO can be used effectively at a Mg/B molar ratio of 20 for three cleaning cycles with 85% removal efficiency, but also notes that an estimated consumption of 18,100 tons of MgO per year makes the process uneconomical in that context. These two statements together frame the typical precipitation tradeoff: precipitation can be effective, but chemical consumption and resulting sludge volumes can become prohibitive when applied to large flows, pushing designers toward polishing methods that transfer boron into smaller secondary streams (e.g., regenerant or concentrate) rather than converting it into bulk solids.

8. Emerging technologies

Several emerging or non-traditional adsorption materials show high reported capacities in the cited review literature, including GO/ZIF-67 with a boron adsorption capacity of 66.65 mg/g at 25 ℃ and nitrogen-doped graphene oxide with a reported maximum adsorption capacity of 58.7 mg/g. Highly variable (and in some cases extremely large) capacities are also reported for other adsorbents (e.g., perlite-GA at 2500 mg/g in the cited review compilation), underscoring that adsorption performance can be strongly material- and test-condition-dependent and must be validated on the target wastewater matrix before adoption.

Electro-driven hybrids are another emerging direction. Boron selective resin-filled EDI is reported to decrease boron from 5 mg/L to 0.3 mg/L with removal up to 94%, with an energy consumption range of 6.64×10^{-3} to 1.11×10^{-1} Wh/g and a reported mass transfer coefficient of 7.78×10^{-4} m/s in that study context. Bipolar membrane electrodialysis (BMED) is reported at pH 9.5–10.5 and current density 6.36 A/m² (max voltage 12 V) with boron removal and recovery efficiencies 98.6% and 86.5%, and an overall energy consumption of 31.5 Wh/g in a cited case, although the corresponding final effluent boron is reported as <10 mg/L (a much less stringent target than <1 mg/L).

9. Recommended treatment train

A pragmatic compliance-oriented approach is to use conventional primary treatment for solids/metals/oils upstream, then add a dedicated boron polishing block designed to guarantee <1 mg/L. The cited literature supports at least three implementable trains.

First, resin polishing (BSR) can serve as the final step because BSR is described as capable of >99% removal, with column operation and regeneration anchored by documented examples (e.g., 4 h^{-1} space velocity and 2 M HCl elution, or NaOH desorption at 0.32 mol/dm^3 achieving 98.5% boron elution in a packed bed example).

Second, multi-pass RO/NF with pH control and/or complexation can be used for boron polishing when concentrate handling is feasible. A two-pass RO with pH adjustment is reported to deliver 99% rejection at feed B = 50 mg/L and pH 10 at 50 $L \cdot m^{-2} \cdot h^{-1}$ flux (SW-30), and staged NF can reduce boron to 0.84 mg/L (3.87⟶0.84 mg/L at pH 10.5 in the cited second-stage example) or even <0.1 mg/L in a two-stage NF with sorbitol complexation and pH 9.8 in the cited case.

Third, hybrid adsorption–membrane systems can be used to "cross the line" below 1 mg/L when adsorption alone stalls above the limit. One demonstrated case shows adsorption (nano-magnetite, 250 min) leaving 2.2 mg/L after 74.39% removal, and subsequent membrane polishing reducing boron to 0.43 mg/L; additionally, a modified (1.0 wt% TiO_2) membrane achieved 94.87% removal to 0.42 mg/L under reported scouring aeration of 5 L/min.

A resin–membrane hybrid is also supported by the cited review: Amberlite IRA743 combined with MF (0.1 μm) is reported at 96–99% removal with resin dosage 3.33 g/L, and an ion exchange (Dowex XUS-43594.00)–UF hybrid is reported to reduce boron concentration to below 1.0 mg/L.

10. Engineering gaps

The literature shows that performance is highly conditional (pH, current density, membrane configuration, and resin cycle time), so the main engineering gap for a bankable magnet-wastewater design is translating "reported-condition" performance into a robust operating window for the actual plant matrix. This uncertainty is explicit in the sources that show large performance swings with chemistry or setpoints, such as RO (~50% at neutral/acidic vs. ~99% at pH 10 under specified conditions), EC removal sensitivity to current density and electrode spacing, and resin cycle time sensitivity to inlet concentration.

Accordingly, a treatment system specification should be built around pilot validation of three quantitative design items that are directly implicated by the provided sources: (1) boron removal vs. pH setpoint in the selected polishing technology (e.g., RO/NF or resin) given the strong pH dependence demonstrated across membrane processes; (2) secondary waste management and mass balance, particularly acid/caustic regenerant requirements for resin trains or concentrate volumes for membrane trains, which are integral to the reported regeneration chemistries and multi-pass membrane approaches; and (3) power and cost sensitivity for electro-driven options, given that EC cost can increase 2.7× with higher current density in one cited case and that EDI/BMED energy consumption is reported over orders of magnitude in Wh/g units across technologies and conditions.

BSR Column Sizing Calculation for Boron Polishing in a 1,000 tpy NdFeB Magnet Facility

Summary

This deliverable sizes a boron-selective resin (BSR) polishing system as a post-treatment step downstream of a conventional metals-precipitation WWTP, using an N-methylglucamine (NMG, "glucamine") chelating resin such as Purolite S108 or Amberlite/AmberSep IRA743 for boron (boric acid/borate) removal. The sizing approach uses (i) a mass load basis (kg B/day), (ii) a resin capacity basis (mol B per bed-liter), and (iii) a hydraulic basis expressed as bed volumes per hour (BV/h), with BV defined as 1 m^3 water per m^3 resin.

Because boric acid is weak (pKa ≈ 9.23) and speciation changes strongly with pH, boron capture performance is typically sensitive to operating pH and competing ions; at higher pH, tetrahydroxyborate becomes dominant and column breakthrough can shift to larger treated volumes (e.g., pH 6⟶11 increased breakthrough volume in a fixed-bed study). The BSR configuration recommended by the resin supplier is a lead–lag (primary + polisher) vessel arrangement, with an optional third equal-sized standby vessel for continuous operation during regeneration.

Design basis

The sizing basis below is an engineering "basis-of-design" example intended to produce a concrete set of column dimensions and operating envelopes; actual final sizing should be validated with site-specific pilot column breakthrough data and vendor modeling (e.g., Purolite's PRSM guidance).

Wastewater flow

For this worked example, the WWTP effluent flow routed to boron polishing is set at **60 m^3/day**, equivalent to **2.5 m^3/h** on a continuous basis, with a **1.5× peak factor** for hydraulics (**3.75 m^3/h**). Bed-volume–based hydraulic calculations use the standard definition **BV = 1 m^3 water per m^3 resin**.

Boron inlet and outlet

The boron concentration entering the BSR system (i.e., after conventional hydroxide precipitation / clarification) is assumed as **C_in = 10 mg/L B**, with a compliance requirement of **C_out < 1.0 mg/L B** and a design target of **C_out < 0.5 mg/L B**. Achieving <0.5 mg/L B is demonstrably feasible with precipitation + adsorption trains in the literature (e.g., 545 mg/L reduced to <0.5 mg/L by precipitation–adsorption).

WWTP effluent matrix

The assumed WWTP effluent matrix for resin protection and performance is neutral to mildly alkaline and low-solids, consistent with the need to avoid premature breakthrough (which is known to be impacted by flow rate and column operating time/throughput in fixed beds). In practical terms, the BSR feed should be filtered (e.g., sand filter or cartridge filtration) such that solids carryover does not cause channeling, pressure drop growth, or resin fouling; fixed-bed modeling and breakthrough monitoring are conventionally used to manage this risk.

Resin selection

NMG-functional resins are designed to remove boric acid/borate from water and brines in packed-bed columns, and are regenerated with dilute mineral acids (HCl or H_2SO_4), followed by caustic conversion steps back to the sodium

form depending on resin type and supplier practice. Purolite states that S108 is an N-glucamine resin that is "largely unimpacted by other ions" and is intended to operate in slightly acidic to slightly alkaline pH, with regeneration via dilute sulfuric or hydrochloric acid, and either co-flow or counter-flow regeneration modes acceptable to achieve low boron leakages (≈0.01–1 ppm B).

For capacity-based sizing (rather than only mg/g lab adsorption numbers), a supplier-reported boron capacity for Purolite S108 is **0.35 mol B per bed-liter** (0.35 mol·L_bed^{-1}). This molar capacity is particularly useful for full-scale sizing because it is already normalized to bed volume (L of resin), matching the BV concept used in ion exchange engineering.

Column sizing

Operating capacity calculation

The supplier-reported boron capacity is taken as **q_total = 0.35 mol B/L_bed** for S108. Converting to a mass basis using boron atomic mass (10.81 g/mol) gives:

- **q_total = 0.35 mol/L × 10.81 g/mol = 3.78 g B/L_bed** (≈3.78 kg B/m^3 resin)

To translate total capacity into a conservative design operating capacity at breakthrough, an engineer typically applies utilization and safety factors (e.g., to account for mass-transfer-zone length, competing ions, resin aging, and conservative switching of lead–lag beds); fixed-bed studies show capacity can decline with regeneration cycling and/or co-solutes (e.g., capacity range during five regenerations; silicon competition reducing boron capacity). In this worked example, we apply a combined conservative utilization factor of **0.68** (0.8 safety × 0.85 competing-ion derate), yielding:

- **q_design = 0.35 mol/L × 0.68 = 0.238 mol/L_bed**
- **q_design = 0.238 mol/L × 10.81 g/mol = 2.57 g B/L_bed** (≈2.57 kg B/m^3 resin)

Boron mass load

The boron mass removal duty is based on the daily treated volume and the concentration reduction required.

With **Q = 60 m^3/day = 60,000 L/day**, **C_in = 10 mg/L**, and **C_out,target = 0.5 mg/L**, the boron mass removed per day is:

- **m_B,removed/day = 60,000 L/day × (10 − 0.5) mg/L × 10^{-3} = 0.57 kg/day**

Resin volume required

Using the conservative design operating capacity **q_design = 2.57 kg B/m^3 resin**, the resin volume required to carry **one day** of boron load to breakthrough-equivalent capacity is:

- **V_resin,1day = 0.57 kg/day ÷ 2.57 kg/m^3 = 0.22 m^3 resin**

In practice, a lead–lag system is sized for a multi-day service run to reduce regeneration frequency and operational risk, and a third standby vessel is used when continuous operation is required during regeneration (merry-go-round operation described by Purolite). For this worked example, set a **7-day** target primary-bed service cycle for the lead vessel before it is removed for regeneration, while the polisher becomes the new lead and the standby becomes the new polisher.

The required resin per vessel (lead capacity basis) is then:

- **$V_{resin,vessel} = 7 \times 0.22 = 1.54\ m^3$** (round to **$1.6\ m^3$** for fabrication and margin)

Service flow rate and EBCT

Ion exchange hydraulic loading is typically described as **space velocity in BV/h**; DuPont's operating conditions guidance lists a **specific flowrate range of 6–60 BV/h** for ion exchange beds in water treatment. Separately, an example column adsorption study for a boron-selective resin reports effective removal at a **space velocity of 4 h^{-1}** (≈4 BV/h), which is a common "conservative" service velocity for specialty polishing where leakage targets are tight.

For the selected vessel resin volume **$V_{resin} = 1.6\ m^3$**, the average and peak service velocities are:

- **$SV_{avg} = Q_{avg} / V = 2.5 / 1.6 = 1.56$ BV/h**
- **$SV_{peak} = Q_{peak} / V = 3.75 / 1.6 = 2.34$ BV/h**

The corresponding empty bed contact times (EBCT) are:

- **$EBCT_{avg} = 60 / SV_{avg} = 38.5$ min**
- **$EBCT_{peak} = 60 / SV_{peak} = 25.6$ min**

These EBCTs fall within a conservative adsorption-contact regime; for adsorption systems, USACE guidance notes that achieving very deep polishing (e.g., from ppm to ppb) can require minimum EBCT on the order of 15–30 minutes, depending on target and matrix.

Column dimensions

DuPont guidance indicates typical single-bed depths (packed-bed systems) in the range **1400–4000 mm**. Selecting a **bed depth H = 2.5 m** (within this range) provides robust hydraulics and mass transfer length for a polishing duty.

For **$V_{bed} = 1.6\ m^3$** and **$H = 2.5$ m**, the cross-sectional area is **$A = V/H = 0.64\ m^2$**, giving an internal diameter:

- **$D = \sqrt{(4A/\pi)} = \sqrt{(4 \times 0.64/\pi)} = 0.90$ m**

This yields an approximate **$L/D = 2.5/0.90 = 2.8$**, consistent with common packed-bed proportions (and within practical internals/distributor design limits).

Hydraulically, the superficial (linear) velocity at peak flow is:

- **$v_{peak} = Q_{peak}/A = 3.75\ m^3/h \div 0.64\ m^2 = 5.9$ m/h**

Number of columns and configuration

A **lead–lag primary + polisher** configuration is the supplier's "normal design" to achieve the lowest leakage when properly operated. For continuous operation during regeneration, Purolite recommends a **third equal-sized vessel** in standby and "merry-go-round" switching logic (lead removed for regeneration; polisher becomes lead; standby becomes polisher).

Accordingly, this sizing example specifies **3 vessels total**, each **$1.6\ m^3$ resin**, configured as:

- **1× lead (primary)** + **1× lag (polisher)** in service, and **1× standby** available for immediate substitution during regeneration

Cycle time

For a simplified capacity-based estimate of lead-vessel run length, use the design capacity per vessel divided by the boron mass load rate. The boron load rate into the BSR (mass basis) is:

- **$m_{B,in}$/day = Q × C_{in} = 60 m^3/day × 10 g/m^3 = 0.60 kg/day**

The vessel's conservative design boron holding capacity is:

- **$M_{B,cap}$ = V_{resin} × q_{design} = 1.6 m^3 × 2.57 kg/m^3 = 4.11 kg B**

A nominal run time to full design capacity is therefore:

- **$t_{service}$ ≈ 4.11 kg ÷ 0.60 kg/day ≈ 6.8 days**

This is consistent with the selected 7-day operating target and supports the 3-vessel merry-go-round scheme for continuous availability.

Regeneration protocol

Regeneration for boron-selective resins is commonly conducted with mineral acid elution (HCl or H_2SO_4) followed by rinsing, with some systems also requiring caustic conversion to sodium form; Purolite explicitly describes dilute acid regeneration and notes both co-flow and counter-flow modes can be used to achieve low boron leakage, while resin aging can increase rinse time requirements. DuPont's operating guidance for ion exchange systems provides typical regeneration concentrations (e.g., **HCl ~5%**) and rinse volumes expressed in BV (e.g., displacement rinse and final rinse ranges), and notes that **regenerant volume preferably ≥ 1 BV** and **flowrate ≥ 2 BV/h** for distribution and contact.

Acid regeneration

For this worked example, specify **HCl elution** as the primary regenerant because fixed-bed boron recovery studies report effective desorption with hydrochloric acid (e.g., **1 M HCl** used for desorption in NMG sorbent columns; **2 M HCl** reported as an effective eluent in brine tests). A conservative, vendor-aligned alternative is **~5 wt% HCl** (DuPont guidance), which is near **1.4 M** and is commonly used in IX practice.

A practical regeneration recipe consistent with DuPont volume guidance is:

- Backwash: **15–30 min** (as needed for solids removal / bed reclassification)
- Acid elution: **1 BV** minimum (often 1–2 BV), at **≥2 BV/h** to ensure distribution, with **contact time >20 min** (but <60 min for sulfuric acid cases)

Rinse

Rinse volumes are expressed in BV and can be structured as a displacement (slow) rinse and a final (fast) rinse. DuPont guidance provides typical displacement rinse volumes (e.g., anion beds often around **~3 BV**) and final rinse volumes (commonly **3–8 BV** for anion beds) with rinse flowrates either the same as service or **10–30 BV/h**.

Purolite also describes an optional "very dilute acid" backflush step consisting of **¼ BV of 0.1% acid** introduced upflow to partially neutralize residual caustic and avoid extended rinsing, particularly in larger plants.

Regenerant brine strength

For brine management planning (downstream of this sizing deliverable), published NMG resin column operation data show boron eluate concentrations on the order of **~0.9–2.1 g/L B** for S108/CRB03/CRB05 when eluted with **1 M HCl**. Such concentrations provide a realistic basis for first-cut regenerant tank material selection and downstream brine management, even though site-specific eluate profiles depend strongly on loading and elution volume.

Annual chemical and resin consumption

Annual reagent usage is driven primarily by (i) the number of regeneration cycles per year and (ii) the chosen regenerant and rinse volumes expressed in BV, since BV is directly tied to resin volume (1 m^3 water per m^3 resin). For a 7-day lead-vessel cycle, the system performs approximately:

- **cycles/year ≈ 365/7 ≈ 52 cycles/year**

If each regeneration uses **1 BV** of acid and **(3 + 6) BV = 9 BV** of rinse water (illustrative within DuPont displacement + final rinse ranges), then per regeneration per vessel:

- Acid volume: **1 BV × 1.6 m^3 = 1.6 m^3 regenerant solution**
- Rinse volume: **9 BV × 1.6 m^3 = 14.4 m^3 rinse water**

This yields annual volumes (per regenerated vessel):

- Acid regenerant solution: **1.6 m^3/cycle × 52 ≈ 83 m^3/year**
- Rinse water: **14.4 m^3/cycle × 52 ≈ 749 m^3/year**

Resin life is application-dependent; general IX references indicate multi-year resin lifetimes under proper care (e.g., 3–8 years), which can be translated into an annual replacement fraction for budgeting and spares.

CAPEX estimate

A defensible vendor-quote-based CAPEX requires specific vessel codes, materials, internals, valve automation philosophy, and corrosion allowances; the literature and supplier guidance provided here does not contain enough information to generate a bankable installed-cost estimate. For order-of-magnitude budgeting, adsorption design guidance provides a historical resin system price range of **$8–$35/lb** for polymeric resins (1998 rates), which can be used only as a rough indicator and must be escalated and validated by supplier quotes for S108/IRA743 in the required ionic form and bead size.

OPEX estimate

Operating cost is dominated by regenerant chemicals and management of the spent regenerant/rinse brine. The available sources support that resins are regenerated using dilute sulfuric or hydrochloric acid, and that rinse volumes are multiple bed volumes per regeneration, so the brine volume is inherently material at industrial scale. The numeric cost basis (acid $/t, wastewater treatment $/$m^3$, hazardous disposal $/$m^3$) is not present in the provided sources, so this report does not assert an all-in $/$m^3$ treated OPEX without additional cost sourcing.

Summary sizing table

The table below consolidates the key sizing outputs for the selected 7-day-cycle, 3-vessel lead–lag–standby design, and also highlights which values are derived from published resin capacity and standard BV definitions.

Parameter	Value	Unit
Design flow (average)	2.5	m^3/h
Design flow (peak)	3.75	m^3/h
Inlet boron	10	mg/L
Outlet boron target	0.5	mg/L
Resin type	Purolite S108 (N-glucamine) or AmberSep/Amberlite IRA743 (NMG)	—
Total capacity basis	0.35	mol B/L_bed
Design capacity (after utilization factors)	0.238	mol B/L_bed
Design capacity (mass basis)	2.57	kg B/m^3 resin
Resin volume per vessel	1.6	m^3
Vessel count	3	lead + lag + standby
Service velocity (avg)	1.56	BV/h
Service velocity (peak)	2.34	BV/h
EBCT (avg)	38.5	min
EBCT (peak)	25.6	min
Bed depth selected	2.5	m
Column ID (calculated)	0.90	m
Nominal service cycle	6.8	days (capacity basis)
Regenerant	HCl (1–2 M typical; ~5 wt% common)	—
Minimum regenerant volume	≥1	BV
Displacement rinse	~3	BV (typical anion bed guidance)
Final rinse	3–8	BV (anion bed guidance)

Methods

The sizing method used here is a conservative, design-for-operations approach that combines resin capacity normalization to bed volume with fixed-bed operational concepts (breakthrough, lead–lag staging, and BV/h hydraulics). Breakthrough behavior is known to be affected by throughput time, inlet concentration, and flow rate in fixed beds, and is commonly modeled with Thomas/Yoon–Nelson/Adams–Bohart type approaches in the boron literature; these models underpin the need for pilot testing to confirm the capacity utilization factor used in full-scale design.

Gaps and recommendations

The most critical engineering gap for a final equipment specification is the lack of site-specific breakthrough curves (treated BV at a defined leakage criterion) for the actual WWTP effluent matrix; competing ions can materially reduce effective capacity (e.g., silicon competition decreasing boron adsorption capacity), and capacity can change across regeneration cycles. The recommended next step is therefore a vendor-supported pilot column test program using the actual post-WWTP effluent and the intended resin grade, run in lead–lag mode with breakthrough monitoring (conductivity monitoring is cited as a viable rapid indicator of breakthrough in ion exchange contexts), and with regeneration performed under the intended acid recipe to generate an actual eluate profile and brine volume for downstream brine management design.

BSR Regenerant Brine Management Strategy for a 1,000 tpy NdFeB Magnet Facility

Summary

A boron-selective resin (BSR) polishing step is typically configured as a lead–lag (primary + polisher) system, with an optional third standby vessel for continuous operation, and is regenerated by acid elution per vendor practice. The regeneration step produces an acidic regenerant/eluate plus rinse waters that contain the removed boron at concentrations that can reach the g/L range when 1 M HCl is used for elution, as shown by column operation data on N-methylglucamine sorbents (e.g., Purolite S108, Diaion CRB03, CRB05). In the available column data set, boron in eluate reached approximately 0.9 g/L (S108), 1.2 g/L (CRB03), and 2.1 g/L (CRB05) under 1 M HCl elution, and in another test set reached 1.5 g/L (S108), 2.2 g/L (CRB03), and 1.7 g/L (CRB05).

For a plant-level design, the brine management decision is driven primarily by (i) the total regenerant + rinse volume per regeneration event (in bed volumes, BV), (ii) the extent to which boron is concentrated into an early "high-B cut" of the acid elution, and (iii) the facility's discharge limit for boron, because conventional neutralization/precipitation is not inherently a boron destruction/removal mechanism and thus blending often simply shifts boron back into the main effluent stream. Three practical pathways are evaluated here: (A) boric acid recovery (concentrate + crystallize), (B) off-site disposal as acidic liquid waste, and (C) neutralization and controlled co-treatment/blending into the WWTP hydraulic stream (feasible only if mass loading remains below the site's boron discharge limit).

Brine characterization

Composition

In the sources available for this task, a common and well-documented regeneration/elution condition for boron-chelating systems is hydrochloric acid at 1 M for elution/desorption of boron from N-methylglucamine sorbents, which necessarily produces an acidic eluate stream (i.e., the "brine" for management). Eluate boron concentrations in the cited column operation data reached roughly 0.9–2.2 g/L depending on sorbent type and operating condition, giving an engineering-relevant concentration range for brine handling and potential recovery operations. Where sulfuric acid is used as an alternative regenerant/eluent, the literature includes 0.5 M H_2SO_4 elution for Amberlite IRA-743 in a column context and 5% H_2SO_4 quantitative elution in an industrial wastewater ion-exchange context, both of which similarly imply an acidic brine stream (sulfate-bearing rather than chloride-bearing).

Trace metals in the brine depend on the upstream WWTP performance and on whether the resin sorbs/strips other trace elements; in a desalination-brine boron recovery context, trace elements sorbed by resin during boron recovery were reported as "below 0.1 ppm" after column elution compared to boron at "1000 ppm", indicating that boron can dominate the eluate composition in some applications. For magnet-factory wastewater, the brine may also contain salts generated in upstream neutralization and polishing steps, but quantitative TDS ranges and trace-metal concentrations for NdFeB-factory BSR brines were not found in the provided sources and therefore must be confirmed by site sampling (ICP-OES/ICP-MS + ion chromatography) before final design.

Volume

Ion-exchange operating guidance defines 1 bed volume (BV) as 1 m^3 water per m^3 resin (or 7.5 gallons per ft^3 resin), which is the correct normalization for estimating regenerant and rinse volumes independent of plant scale. For efficient hydraulic distribution, DuPont operating guidance recommends a regenerant solution volume of one bed volume or more, and also recommends that regenerant flowrate not be too low (≥2 BV/h), implying that regenerant

volume is a first-order determinant of brine volume per cycle. The same guidance gives displacement rinse volumes of 2 BV (cation) or 3 BV (anion) and final rinse volumes of 2–4 BV (cation) or 3–8 BV (anion), with rinse flowrate either the same as service flowrate or 10–30 BV/h, which sets the magnitude of rinse-water contributions to "brine" volume if rinses are sent to the brine tank rather than recycled or returned to the equalization tank.

If the BSR system is treated as an "anion-style" regeneration sequence for volume planning purposes (acid elution followed by rinses), then a practical per-regeneration-event brine volume bound can be estimated from cited ranges as: regenerant ≥1 BV plus displacement rinse ≈3 BV plus final rinse ≈3–8 BV, giving approximately 7–12 BV total liquid generated per cycle per m\³ of resin inventory, before any optional intermediate steps (e.g., very dilute acid backflush). The Purolite BSR design guidance also describes an optional additional "very dilute acid step" consisting of 1/4 BV of 0.1% acid introduced as an upflow backflush to partially neutralize residual caustic and reduce extended rinsing for aging resins, which may add to the total brine/rinse volume if that step is not recycled within the process.

Option A

Option A is to treat the BSR brine as a resource stream by maximizing boron concentration in a segregated "high-B" eluate cut and then applying concentration and crystallization to produce boric acid (H\₃BO\₃) or a borate product suitable for sale or internal reuse. The core enabling observation is that acid elution can create a boron concentration factor relative to the treated water, with one reported acidic elution (1 M HCl) achieving concentration factors of 11 for boron (and 35 for germanium) in a column setting, demonstrating that eluate concentration is a controllable design lever rather than a fixed property.

Process sequence

A practical recovery-focused sequence begins with (1) brine collection and equalization (segregated from normal WWTP influent), followed by (2) fractionation of the elution profile to isolate the highest-boron fraction, since column data show a concentrated eluate boron level in the g/L range under 1 M HCl elution and the objective is to maximize that concentration into the minimum volume for downstream evaporative concentration. The brine can then be (3) conditioned for concentration (e.g., remove suspended solids via cartridge filtration if present, and, if necessary, neutralize partially to reduce corrosion risk), after which (4) evaporation concentrates boron to higher levels suitable for crystallization, and (5) cooling crystallization and solid–liquid separation produce a boric-acid crystal product; however, quantitative solubility curves, crystallization yields, and evaporator energy consumption were not present in the provided sources and thus must be sourced from boric-acid property data and vendor design packages during FEED.

Where electrodialysis with bipolar membranes (EDBM/BMED) is available, the literature also supports integrating electro-driven concentration and reagent recovery steps in boron recovery schemes, including an EDBM case with Na and B removal of ~97% after 6 h at 9.5 V and current efficiency for H\₃BO\₃ production of ~80%, which may be relevant as a higher-technology alternative to pure thermal evaporation depending on brine flow and energy price.

Product quality

If the recovery route is implemented via chromatographic separation for boric acid purification (rather than direct crystallization), boric acid purity "99.5–100%" (conforming to Merck KGaA standard grade) has been reported, indicating that high-purity product is achievable in at least some boron-industry wastewater contexts when using ion-exchange/chromatographic separation. A key risk for a magnet-factory brine is trace metal carryover into product, which must be quantified (Ni, Cu, Fe) and controlled using fractionation, polishing, or purge strategies; the available sources indicate that in some brine-valorization column systems, trace elements after elution were below

0.1 ppm while boron was at 1000 ppm, but analogous data for magnet wastewater brines are not available and require pilot testing on real brine to confirm achievable product specifications.

Sizing basis and mass balance

The data set provides enough parameters to construct a conservative brine mass-balance framework per unit resin volume even without site-specific wastewater flow: (i) brine volume per regeneration event of ~7–12 BV per m^3 resin inventory, and (ii) eluate boron concentration in the range ~0.9–2.2 g/L for 1 M HCl elution in the cited column work. On that basis, the boron mass generated per regeneration event per m^3 resin is approximately 6.3–26.4 kg B per cycle (i.e., 7–12 m^3 × 0.9–2.2 kg/m^3), which is the loading that the recovery or disposal system must handle per cycle if all regenerant and rinse streams are routed to brine management rather than partially recycled to the equalization tank.

Economics

No direct CAPEX/OPEX data for small-scale boric-acid recovery by evaporation/crystallization from BSR regenerant brine were present in the provided sources, so any economic conclusion at 1,000 tpy magnet scale is necessarily preliminary and should be replaced with a vendor-quoted estimate after brine sampling and a heat-and-material balance are completed. Where boron recovery is explicitly framed as economically meaningful, one cited study notes the feasibility of recovering boron at scale through brine management and product generation (e.g., via EDBM and integrated recovery schemes), but the quantitative cost basis is not supplied in the quotes provided here and must be obtained in a follow-on cost-estimating step.

Option B

Option B is to treat the BSR brine as a waste stream and ship it off-site for permitted disposal, which is often the fastest path to compliance when the brine volume is small or when the facility has limited space and utility capacity for an evaporator/crystallizer system. From a regulatory-control standpoint, the brine is clearly an "ion exchange regeneration" liquid containing the removed contaminant, and the literature supports using conductivity-based sensing to detect breakthrough and drive regeneration scheduling, which also enables prediction of brine generation frequency for logistics planning.

Quantitative disposal cost ranges (e.g., $/$m^3$), RCRA waste codes, or detailed TSDF unit operations for boron-bearing acidic liquids were not contained in the provided sources, and therefore must be developed from jurisdiction-specific hazardous waste regulations, waste characterization (pH/corrosivity, metals), and hauler/TSDF quotes during project development rather than inferred from the present literature subset.

Option C

Option C is to neutralize the BSR brine (e.g., with NaOH or $Ca(OH)_2$) and return it to the WWTP equalization tank for blending and re-treatment, relying on dilution rather than boron removal as the compliance mechanism. This option is only feasible if the mass of boron returned to the WWTP divided by the total treated effluent volume keeps the final discharge concentration below the permit limit, because the cited literature emphasizes that very low boron effluents (e.g., <0.5 mg/L) are achievable with precipitation+adsorption combinations, implying adsorption (or a similarly selective step) is the true "boron removal" mechanism rather than conventional hydroxide precipitation alone.

Feasibility check

A defensible feasibility check can be done on a boron mass-balance basis per m^3 resin without assuming a specific facility flow: the boron mass returned per regeneration event is approximately (brine volume per cycle) × (average brine boron concentration), where the brine volume per cycle is ~7–12 m^3 per m^3 resin and a representative boron concentration can be taken from the cited eluate range of ~0.9–2.2 g/L under 1 M HCl elution. The incremental boron concentration added to the WWTP effluent by blending is then $\Delta C_B = m_B / V_{effluent}$, where $V_{effluent}$ is the total effluent volume over the blending period, and if ΔC_B alone approaches the discharge limit then Option C is not viable and simply recycles the boron problem back to the BSR inlet. The need for this calculation is reinforced by the observation that boron speciation changes with pH ($pK_a \approx 9.23$), which affects capture on boron-selective media but does not create a precipitation sink by itself, so neutralizing the brine does not eliminate boron mass and typically only changes its speciation and distribution between $B(OH)_3$ and $B(OH)_4^-$.

Comparative analysis

The comparison below summarizes the three options using only attributes supported by the provided sources and flags where quantitative cost data are missing.

Criterion	Option A Recovery	Option B Disposal	Option C Co-treatment
Core mechanism	Concentrate boron via acid elution (CF reported ~11) then recover as product (route-specific)	Ship acidic regenerant + rinses off-site; regeneration volume typically ≥1 BV plus rinses	Neutralize and dilute back into WWTP; boron mass conserved so feasibility depends on mass balance vs discharge limit
Brine B level supported by data	0.9–2.2 g/L B reported in 1 M HCl eluate for NMG sorbents	Same as generated by regeneration/elution chemistry and fractionation choices	Same as generated; returned to equalization tank after neutralization
Brine volume supported by data	~7–12 BV per cycle per m^3 resin (regenerant ≥1 BV + 3 BV displacement + 3–8 BV final rinse)	Same volume basis; depends on whether rinse streams are segregated or recycled	Same volume basis; depends on equalization tank capacity and WWTP hydraulic load
Technical risk	Medium due to need for solubility/energy design data not present here; may integrate EDBM (97% Na and B removal after 6 h at 9.5 V; ~80% current efficiency for H_3BO_3)	Low technical risk at plant; higher ongoing logistics dependency; costs not in provided sources	High compliance risk if blended boron exceeds permit limit; must be validated by mass-balance and monitoring of breakthrough/leakage
Sustainability	Highest if product recovery is feasible; high-purity boric acid 99.5–100% has been demonstrated in boric-acid recovery contexts	Lowest because it externalizes the waste burden; quantitative metrics not provided	Moderate; reduces off-site disposal but may be infeasible if discharge limit is stringent

Criterion	Option A Recovery	Option B Disposal	Option C Co-treatment
CAPEX and OPEX evidence in provided sources	Not quantified in provided sources; requires FEED-level design and vendor quotes	Not quantified in provided sources; requires TSDF pricing and transport quotes	Not quantified; depends on neutralization chemical use and WWTP flow; neutralization concentration guidance exists for IX systems but not specific to BSR brine costs

Recommendation

For a 1,000 tpy NdFeB facility, the available sources support treating the BSR regenerant as a g/L-boron acidic stream produced by 1 M HCl elution (or comparable acid strengths), but they do not provide enough cost and property data (e.g., boric acid solubility and crystallization energetics) to justify recovery economics without additional site-specific testing and vendor engagement. Accordingly, a conservative near-term recommendation is to implement Option B (off-site disposal) as the compliance baseline while running a brine segregation/fractionation pilot (early high-B cut vs rinse) to generate the data needed to evaluate Option A at a higher-confidence level, including achievable concentration factor, trace metal carryover, and potential product purity. Option C should be treated as a contingency only after a quantitative mass-balance check demonstrates that boron mass returned to the WWTP will not drive final effluent above the permit limit, because boron removal to sub-mg/L levels is explicitly demonstrated when adsorption is included (e.g., from 545 mg/L to <0.5 mg/L via precipitation–adsorption combination), indicating that relying on dilution alone is generally a fragile compliance strategy.

Integration

The brine management system must be physically integrated with the BSR skid through dedicated tankage, valves, and instrumentation that supports brine segregation and automated fractionation during regeneration, with operations governed by a predictable regeneration sequence and switching logic for lead–lag vessels. At minimum, integration should include a regenerant preparation tank, a brine collection/equalization tank sized on BV-based cycle volumes, and routing that allows (i) high-B eluate to be isolated for Option A/B and (ii) selected low-boron rinses to be recycled or returned to equalization when permitted by the WWTP mass balance and discharge limit. Breakthrough detection and regeneration timing can be supported by online conductivity monitoring, which has been proposed as a practical control parameter for detecting breakthrough in boron ion-exchange operations and thus can be used to stabilize brine generation frequency for logistics and downstream equipment scheduling.

Engineering-Level Technical Review of Permanent Magnet Manufacturing Waste

1. SUMMARY

This technical review provides an engineering-level analysis of the waste streams generated during the manufacturing of sintered Neodymium-Iron-Boron (NdFeB) and Samarium-Cobalt (SmCo) permanent magnets. For a typical facility producing 1,000 to 10,000 tonnes per year, manufacturing waste can account for up to 30-50% of the initial raw material input, with the machining stage being the single largest contributor. This stage generates a high-volume, hazardous grinding sludge containing 20-30% rare earth elements (REEs) and 60-70% iron, suspended in water- or oil-based coolants. An environmental assessment for a proposed U.S. facility estimates weekly generation of 16,600 kg of hazardous rare-earth grinding slurry and 25 m^3 of spent coolant.

Other significant waste streams include airborne fines from powder handling, off-specification rejects from pressing and sintering, and contaminated wastewater from surface treatment and coating operations. Surface treatment, particularly Ni-Cu-Ni electroplating, generates acidic and alkaline rinse waters containing dissolved metals and spent plating bath solutions requiring specialized treatment.

Current treatment predominantly relies on conventional chemical precipitation for wastewater, where pH is adjusted (typically to 9.0-9.5) to precipitate metal hydroxides, followed by coagulation, flocculation, clarification, and sludge dewatering via filter press. The resulting dewatered sludge is classified as hazardous waste and incurs significant disposal costs, often representing the largest single operational expenditure in waste management. Disposal costs for hazardous sludge can range from $220 to over $22,000 per tonne depending on location and composition.

Regulatory frameworks, such as the US EPA's 40 CFR 433 (Metal Finishing) and the EU's Best Available Techniques (BAT) Reference Documents (BREFs), impose strict numeric discharge limits for metals like nickel (2.38 mg/L monthly avg, US EPA) and copper (2.07 mg/L monthly avg, US EPA). The high concentration of valuable REEs in machining sludge (up to 30 wt%) presents a significant economic opportunity, though commercial-scale recovery from manufacturing waste remains limited, with compliance-driven disposal being the primary driver of current practices.

2. WASTE STREAMS INVENTORY

The manufacturing of sintered permanent magnets is a multi-stage process where waste is generated at every step. The following inventory details the primary waste streams from a facility with a sintered magnet output of 1,000-10,000 tonnes/year.

A. POWDER HANDLING AND PRESSING

This initial stage involves milling the cast alloy into a fine powder, followed by pressing it into ”green” compacts.

- **Waste Type**: Waste streams include oxidized powder, off-spec powder, press rejects, and airborne fines collected by dust extraction systems.
- **Composition**: The powder consists of Nd2Fe14B alloy particles, Nd-rich grain boundary phases, and surface oxides (Nd2O3, Fe2O3). Jet-milled powders typically have a narrow particle size distribution around 3-5 µm.
- **Volume Generated**: Yield loss at this stage can be significant, with one source suggesting up to 5% of the magnetic alloy ends up as fine powder waste. A proposed facility estimated generating 500 kg/week of oxidized powder and green body residues.
- **Hazard Classification**: Fine NdFeB powder is pyrophoric and can ignite spontaneously in air. Safety Data Sheets (SDS) classify it as a flammable solid and recommend handling in inert atmospheres (nitrogen, argon)

or as a water slurry to prevent combustion. Respirable cobalt dust (grain size ≤10μm) is classified as acutely toxic by inhalation.

B. SINTERING AND HEAT TREATMENT

The green compacts are sintered at high temperatures in a vacuum or inert atmosphere to densify the material.

- **Waste Type**: Furnace residues, off-specification magnets that fail quality control (density, dimensions, magnetic properties), and contaminated vacuum pump oil are the primary wastes.
- **Sintering Parameters**: Sintering occurs at temperatures between 1000-1100℃, followed by two-stage annealing (e.g., 850-950℃ and 400-600℃). During this process, magnets can shrink anisotropically by 7-25% linearly.
- **Rejects**: Magnets that fail post-sintering quality checks for density (>98% of theoretical) or magnetic properties are rejected. These rejects can be re-processed via hydrogen decrepitation to reclaim the alloy powder or are scrapped.

C. MACHINING (CUTTING, GRINDING, LAPPING)

This is the primary waste-generating stage, where sintered blocks are machined to precise final dimensions. Overall material loss in this stage can be between 20-30% of the sintered block weight. A large portion of this waste, over 70%, is in the form of sludge.

- **Grinding Sludge/Slurry**:
  - **Carrier Fluid**: Water-based emulsions are commonly used as coolants. One common type is a mineral oil and boric acid-based, water-miscible fluid (e.g., Mobilcut 230) used at a 5% concentration with a throughput of 30–50 L/min per grinding wheel. These fluids typically contain emulsifiers, biocides, corrosion inhibitors, and pH buffers, operating at a pH of 9.2-10.0.
  - **Metal Content**: Dried sludge solids contain high concentrations of magnet materials. Compositions vary, but typical ranges are 20-30 wt% REEs (Nd, Pr, Dy) and 60-70 wt% Iron. An exemplary grinding slurry composition is Nd 19.5%, Pr 4.9%, Dy 2.4%, B 0.8%, Co 1.3%, Al 0.2%, and Fe as the balance.
  - **Abrasive Particles**: Diamond abrasives are commonly used for grinding NdFeB magnets.
  - **Sludge Properties**: The density of grinding slurry can range from 3150 to 4075 kg/m^3. The moisture content of as-generated slurry is high, and dewatered cake solids can vary. A proposed facility estimated generating 16,600 kg/week of hazardous grinding sludge.
- **Wire EDM Waste**:
  - **Materials**: This process uses a thin (0.25 mm) brass (CuZn37) or coated wire as an electrode. The dielectric fluid is typically deionized water or oil.
  - **Waste**: The waste consists of metal particles from the workpiece and wire eroded into the dielectric fluid, which is then filtered. The filter cake contains a mixture of these metals. Kerf loss is minimized by using very fine wire, but a thermally affected, demagnetized layer of approximately 35 μm can be formed on the cut surface.
- **Total Material Loss**: The fraction of the sintered block lost as waste during machining varies by final geometry. SmCo magnets are particularly brittle and can have machining failure rates as high as 20%.

D. SURFACE TREATMENT AND COATING

To prevent corrosion, machined magnets undergo cleaning and coating.

- **Pre-treatment**:
  - **Alkaline Degreasing**: Baths typically contain NaOH and surfactants, operated at elevated temperatures.

  - **Acid Pickling/Etching**: Nitric acid (HNO3) at 1-10% mass fraction is often used to remove surface oxides.
  - **Rinse Water**: Multiple rinse stages generate large volumes of wastewater containing dragged-out cleaning solutions and dissolved metals. Water usage is estimated at 3-20 L/m² of surface area per rinse stage.
- **Electroplating (Ni-Cu-Ni)**: This is the most common coating.
  - **Watts Nickel Bath**: A typical bath contains Nickel Sulfate (NiSO4·6H2O) at 240-300 g/L, Nickel Chloride (NiCl2·6H2O) at 30-90 g/L, and Boric Acid (H3BO3) at 30-45 g/L. It operates at 40-60℃, a pH of 3.5-4.5, and a current density of 2-7 A/dm².
  - **Copper Bath**: Acidic copper baths typically contain copper sulfate (CuSO4) and sulfuric acid (H2SO4).
  - **Wastewater**: Drag-out from plating baths contaminates subsequent rinse waters with Ni and Cu. Bath maintenance generates sludge from carbon treatment and anode dissolution.
- **Alternative Coatings**: Other coatings include epoxy, phosphate, and aluminum sputtering (IVD), each generating unique waste streams such as solvents, phosphate sludge, or chamber cleaning waste.
- **Total Wastewater**: A facility producing 1,000 tonnes/year can generate a significant volume of wastewater from surface treatment. A proposed US facility anticipated discharging 4,755 gallons per day (18 m³/day) of process water.

E. QUALITY REJECTION AND PACKAGING

Final inspection leads to the rejection of magnets that fail dimensional or magnetic tests. A proposed facility estimated generating 45 barrels per week of off-spec magnets, which would be recycled internally.

3. WASTE CHARACTERIZATION AND HAZARD CLASSIFICATION

Each waste stream possesses distinct characteristics and hazard classifications critical for determining appropriate treatment and disposal pathways.

- **Pyrophoricity of Powders**: Fine NdFeB powders are pyrophoric and classified as flammable solids. They can spontaneously ignite on contact with air, especially at smaller particle sizes, and react with water to generate flammable hydrogen gas. They should be stored under an inert atmosphere or as a water slurry.
- **Machining Sludge**:
  - **RCRA Classification**: Grinding sludge is typically classified as a hazardous waste in the US due to the leachable toxicity of heavy metals (e.g., Ni, Co). A proposed US facility classifies its 16,600 kg/week of rare-earth sludge/grinding slurry as hazardous.
  - **Composition**: A typical grinding slurry may contain 19.5% Nd, 2.4% Dy, 4.9% Pr, 0.8% B, 1.3% Co, and ~5.6% oxygen in the solid phase, with the balance being iron. The oil content in sludge from oil-based coolants can be as high as 51%.
- **Surface Treatment Wastes**:
  - **Spent Acids/Alkalines**: These are corrosive hazardous wastes.
  - **Plating Sludge**: Sludge from wastewater treatment contains high concentrations of metal hydroxides (e.g., Ni(OH)2, Cu(OH)2, Fe(OH)3) and is classified as hazardous waste. An example sludge contained 198 g/kg zinc and 129 g/kg iron (dry basis) with a moisture content of 60%.
- **Combined Wastewater Characterization**: The combined effluent from a magnet manufacturing facility before treatment is typically characterized by:
  - **pH**: Varies widely due to batches of acidic and alkaline discharges.
  - **Total Suspended Solids (TSS)**: High, primarily from machining fines and precipitated solids.
  - **Chemical Oxygen Demand (COD)**: High, due to organic components from coolants, cleaners, and plating additives.

- **Oil and Grease**: Significant if oil-based coolants are used.
- **Total Metals**: Elevated concentrations of Nd, Dy, Fe, B, Ni, and Cu.

4. TREATMENT PROCESSES — STEP-BY-STEP ENGINEERING DETAIL

The following sections detail the current standard treatment processes for the major waste streams in a permanent magnet manufacturing facility.

A. WASTEWATER TREATMENT PLANT

A conventional chemical-physical wastewater treatment plant (WWTP) for a magnet facility follows a multi-step sequence.

- **Step 1: Oil/Water Separation**: For facilities using oil-based coolants, initial treatment involves oil removal. A common method is an oil/water separator or a Dissolved Air Flotation (DAF) unit to skim free and emulsified oils.
- **Step 2: pH Adjustment & Precipitation**: The wastewater pH is raised to between 9.0 and 11.0 to precipitate dissolved metals as hydroxides. Lime (Ca(OH)2) or caustic soda (NaOH) are common reagents. For REEs, optimal precipitation occurs at a higher pH, often requiring a two-stage process for effective removal of both REEs and other metals like nickel and copper.
- **Step 3: Coagulation/Flocculation**: A coagulant such as ferric chloride (FeCl3) or polyaluminium chloride (PAC) is added to neutralize charged particles. This is followed by the addition of a polymer flocculant to agglomerate the small precipitated particles into larger, settleable flocs.
- **Step 4: Clarification/Settling**: The flocculated water flows into a clarifier (e.g., inclined plate or circular) where the solids settle out as sludge. The clarified water overflows for further treatment or discharge.
- **Step 5: Sludge Dewatering**: The sludge from the clarifier, typically with a low solids content (1-5%), is pumped to a dewatering unit, most commonly a filter press. The press forces water out of the sludge, producing a "filter cake" with a much higher solids concentration (typically 30-55% for metal hydroxide sludge). Filter presses for this application operate at pressures of 7 to 16 bar (100 to 225 psi).
- **Step 6: Polishing**: If required to meet stringent discharge limits, the clarified water may undergo polishing steps such as sand filtration, activated carbon adsorption (for organics), or ion exchange (for trace metals).

B. SLUDGE MANAGEMENT

The dewatered filter cake is the largest hazardous solid waste stream by volume.

- **Volume and Composition**: A facility producing 1,000 tonnes/year of magnets could generate hundreds of tonnes of dry solids in sludge annually. The composition reflects the magnet alloy and any plating metals, consisting primarily of rare earth and iron oxides/hydroxides.
- **Disposal**: The filter cake is typically transported to a certified hazardous waste landfill. Disposal costs are a major operational expense, varying by region but often in the range of several hundred USD per tonne.
- **Recycling**: While the sludge contains valuable REEs (20-30 wt%), commercial-scale recovery from manufacturing sludge is not yet widespread. The primary driver for sludge management remains environmental compliance and cost of disposal rather than value recovery.

C. COOLANT MANAGEMENT

Coolant lifecycle involves preparation, in-use monitoring for concentration, pH, and bacterial count, and eventual disposal.

- **Treatment**: Spent coolant emulsions are "cracked" using acid, salt, or heat to separate the oil and water phases. The oil phase is typically incinerated, and the aqueous phase is sent to the WWTP for treatment of dissolved metals and organic components.
- **Volume and Cost**: A proposed facility estimated generating 25 m^3/week of spent coolant, which is disposed of as hazardous waste. Disposal costs are typically charged per cubic meter.
- **Recycling**: On-site coolant recycling systems using centrifuges or membrane filtration can extend coolant life and reduce disposal volumes, but require capital investment.

D. PLATING BATH AND RINSE WATER TREATMENT

- **Spent Baths**: Concentrated spent plating solutions are either treated on-site or hauled off-site as hazardous waste. Nickel can be recovered from spent Watts baths through electrolytic recovery, with an energy consumption of about 3.1 kWh/kg of Ni recovered.
- **Rinse Water**: Drag-out from plating tanks contaminates rinse waters. To minimize water usage and waste, multi-stage counter-current rinsing is a BAT. Ion exchange systems can be used on rinse waters to recover Ni and Cu, allowing the water to be recycled and the recovered metals to be returned to the plating bath.

5. COSTS AND ECONOMICS

For a reference facility producing 1,000 tonnes/year of sintered NdFeB magnets, the costs of waste management are substantial.

A. Waste Generation Volumes

- **Total Machining Waste**: Assuming a 25% material loss during machining, a 1,000 tpy facility would generate approximately 250 tonnes of swarf and sludge solids annually.
- **Total Wastewater**: Based on a new US facility's projections, a similar-sized plant might generate around 4,755 gallons/day (approx. 6,600 m^3/year) of process wastewater.
- **Spent Coolant**: The same facility projects 25 m^3/week, equating to 1,300 m^3/year of spent coolant.

B. Treatment Costs (CAPEX)

- **Wastewater Treatment Plant**: For a flow rate of ~20-30 m^3/day, a complete chemical precipitation and sludge dewatering system can have a CAPEX ranging from $400,000 to over $1.5 million, depending on complexity and automation. Major equipment includes clarifiers ($40,000 - $150,000), filter presses, chemical dosing systems, tanks, and pumps.

C. Treatment Costs (OPEX)

- **Chemicals**: Costs for lime/caustic, coagulants, and polymers can be significant. A 5,000 m^3/day plant using a cerium-based process estimated annual chemical costs exceeding $1.5 million, with HCl being a major component.
- **Sludge Disposal**: This is often the largest single OPEX item. At a generation rate of 250 dry tonnes/year, and a disposal cost of $300-$1000/tonne, annual costs would be $75,000 - $250,000.

- **Labor and Maintenance**: Operating a WWTP requires dedicated staff. Maintenance is typically estimated at 3-5% of CAPEX annually.

D. Total Cost of Waste Management

The total annual cost of waste management can easily exceed several hundred thousand dollars, representing a significant cost per tonne of magnet produced. Sludge disposal and chemical consumption are typically the dominant cost drivers.

E. Value of Metals in Waste

- **REE Content in Sludge**: With a sludge generation of 250 tonnes/year (dry) and an average REE content of 25%, the sludge contains approximately 62.5 tonnes of REEs.
- **Market Value**: The market value of these REEs (primarily Nd, Pr, and Dy) can be substantial, often exceeding the cost of disposal. However, the economic viability of recovery depends on the capital and operating costs of a dedicated hydrometallurgical or pyrometallurgical recovery plant, which are currently high for on-site implementation at this scale.

6. REGULATORY REQUIREMENTS — SPECIFIC NUMERIC LIMITS

Wastewater discharges from magnet manufacturing facilities are regulated under metal finishing and nonferrous metals processing guidelines.

Pollutant	US EPA 40 CFR 433 (mg/L)	EU IED (FMP BREF) (mg/L)	China GB 25467-2010 (mg/L)
	Monthly Average	**Yearly Average**	**Daily Maximum**
Copper (Cu)	2.07	0.005–0.2	0.5
Nickel (Ni)	2.38	0.01–0.2	0.5
Zinc (Zn)	1.48	0.05–1	1.5
Iron (Fe)	-	1–5	5.0
Total REE	-	-	0.5
Boron (B)	-	-	-
TSS	31	5–30	30
COD	-	30–90	100
pH	6.0–9.0	6.0–9.5	6.0–9.0

Note: EU limits are BAT-AELs for direct discharge from the Ferrous Metals Processing (FMP) BREF, as a proxy. NFM and STM BREFs provide similar ranges.

The most difficult contaminants to manage are often the rare earth elements themselves, as they are not specifically regulated in all jurisdictions and their precipitation chemistry can be complex, especially in mixed-metal waste streams. Achieving very low levels of multiple metals simultaneously requires precise pH control, often in multiple stages.

7. METAL MASS BALANCE

For a reference 1,000 tonnes/year NdFeB facility, a simplified mass balance highlights material losses. Assuming a typical alloy composition of 27% Nd, 4% Pr, 1% Dy, 67% Fe, and 1% B:

- **Total Input**: ~1,430 tonnes/year of raw alloy (assuming a 70% overall yield).
  - Nd: ~386 tonnes
  - Pr: ~57 tonnes
  - Dy: ~14 tonnes
  - Fe: ~958 tonnes
  - B: ~14 tonnes
- **Finished Magnets**: 1,000 tonnes/year.
  - Nd: ~270 tonnes
  - Pr: ~40 tonnes
  - Dy: ~10 tonnes
  - Fe: ~670 tonnes
  - B: ~10 tonnes
- **Total Waste Losses**: ~430 tonnes/year.
  - **Powder/Pressing Loss (~5%)**: ~21.5 tonnes (primarily alloy powder)
  - **Sintering Rejects (~5%)**: ~21.5 tonnes (sintered blocks)
  - **Machining Loss (~20%)**: ~86 tonnes (as dry solids in sludge/swarf)
    - Contains ~21.5 tonnes Nd, ~3.4 tonnes Pr, ~0.9 tonnes Dy
  - **Surface Treatment Loss**: Minor mass loss from etching, but contributes dissolved metals to wastewater.

This estimation indicates that approximately 116 tonnes of Nd, 17 tonnes of Pr, and 4 tonnes of Dy are lost to various waste streams annually for every 1,000 tonnes of magnets produced.

8. ENGINEERING GAPS AND RECOMMENDATIONS

A significant gap in the publicly available literature is the lack of comprehensive, facility-scale operational data that links waste volumes, compositions, and treatment costs directly to magnet production tonnage. Most data comes from lab-scale recycling studies or environmental assessments with estimated figures. To design an advanced or alternative treatment system, an engineer would require:

- Detailed, time-series data on the flow rates and compositions of individual wastewater streams (e.g., grinder effluent, plating rinses, acid pickles) to optimize segregated treatment.
- Particle size distribution and settling characteristics of machining sludge to better design clarifiers and dewatering equipment.
- Leachability data (e.g., TCLP) for dewatered sludge to assess disposal options and recycling potential.
- Pilot-scale testing data for emerging technologies on actual magnet manufacturing wastewater.

Emerging Technologies:

- **Membrane Filtration (RO/NF)**: Reverse osmosis and nanofiltration can achieve very high-quality effluent for water reuse and can concentrate dissolved metals, but they face challenges with membrane fouling from oils and precipitated solids. Costs are higher, with specific energy consumption for RO potentially reaching 20.5 kWh/m^3.
- **Selective Adsorbents**: Functionalized resins or chelating materials could selectively target and recover high-value REEs from dilute rinse waters, but their capacity, regeneration, and cost-effectiveness in complex indus-

trial streams need further validation.

- **Zero-Liquid-Discharge (ZLD)**: ZLD systems, typically using evaporators and crystallizers, can eliminate wastewater discharge entirely but are highly energy-intensive and have a high CAPEX. A 1-20 GPM ZLD system can cost between $250,000 and $2 million.
- **Biological Treatment**: While not common for the high-metal, low-BOD wastewater from magnet manufacturing, biosorption using algae or other biomass has been studied for REE recovery at lab scale, with reported adsorption capacities of 25-65 mg/g for lanthanum. Commercial application is not yet established.

Integrated WWTP, Boron Polishing, and REE Recovery PFD for a 1,000 tpy Sintered NdFeB Magnet Facility

Summary

This report provides an integrated, facility-level Process Flow Diagram (PFD) narrative that connects three treatment systems: (A) a central wastewater treatment plant (WWTP) for metals- and oil-bearing aqueous wastes, (B) a boron-selective resin (BSR) polishing system to achieve sub-mg/L boron in final effluent, and (C) a separate hydrometallurgical rare-earth element (REE) recovery plant that processes NdFeB machining/grinding sludge as a solids stream (not routed through the WWTP). The integration logic is driven by boron aqueous chemistry (boric acid/borate speciation) and by the very high REE/Fe content of machining sludges (often ~20–30% REEs and 60–70% Fe) that makes separate solids routing and higher-value recovery practical.

Boron is difficult to remove by conventional neutralization because boron exists primarily as weakly dissociated boric acid at typical neutral pH; speciation shifts near pH ≈ 9.23 (pKa), which is also why many boron-removal technologies either use boron-selective complexation media (e.g., N-methylglucamine functional groups) or drive speciation via high-pH membrane approaches.

Methods

The boron-polishing portion of the integrated design is grounded in fixed-bed column adsorption/ion exchange literature and vendor design guidance that explicitly uses breakthrough-curve modeling (Thomas and Yoon–Nelson models) and lead–lag vessel operation to control leakage.

The REE recovery portion of the integrated design uses published NdFeB sludge and magnet-waste hydrometallurgical routes that quantify roasting conditions (including 800 ℃ and 2 h with controlled airflow), stirred-tank leaching conditions (e.g., 300 mL reactor at 300 rpm and 80 ℃), precipitation stoichiometry for oxalate recovery, and post-precipitation calcination to oxide.

Process overview

Three-system architecture

System A is a central WWTP whose function is to accept mixed aqueous wastes and produce clarified effluent suitable for downstream boron polishing. Because the provided dataset for this integrated deliverable is dominated by boron-removal and REE-recovery sources (rather than metal-finishing ELG/BREF WWTP design texts), this section describes the WWTP as an interface-and-segregation node and explicitly flags unit-operation parameters that require site-specific characterization for a bankable design.

System B is a BSR polishing system using N-glucamine/N-methylglucamine functionalized media (e.g., Purolite S108) designed for operation in slightly acidic to slightly alkaline pH and regenerated with dilute sulfuric or hydrochloric acid; vendor guidance emphasizes lead–lag vessels (primary + polisher) and an optional third standby vessel for continuous operation.

System C is an REE recovery plant that receives machining/grinding sludge (as a solids stream) and applies thermal pre-treatment and hydrometallurgy to recover REEs and manage iron either by selectivity (oxidative roasting + selective leach) or by co-leaching and stepwise precipitation; literature reports that NdFeB sludges can contain roughly 20–30% REEs and 60–70% Fe.

Key interconnections

The integrated facility interconnections are as follows.

First, WWTP clarified effluent is routed to the BSR system because BSR can "almost remove all boron" and can reach removal rates above 99%, enabling a polishing barrier after conventional metals precipitation and solids removal.

Second, BSR regeneration generates a concentrated boron-bearing regenerant (brine/eluate), which must be managed separately; published work demonstrates that boric acid can be recovered to high purity (99.5–100%) using separation approaches, and that B can be concentrated and recovered at high yield in column-elution approaches (e.g., concentration factor 21 with 98% recovery in one brine-focused study).

Third, REE recovery plant liquid effluents (acidic raffinate/filtrates/washes) are returned to the WWTP equalization and metals-precipitation front end because REE-recovery routes explicitly involve strong acids (HCl or H2SO4), chloride media (e.g., 4 M Cl− solvent-extraction feed matrices), and neutralization/precipitation steps that generate aqueous side streams requiring downstream treatment.

Fourth, machining/grinding sludge is routed directly to the REE recovery plant, not through the WWTP, because it is a concentrated solids resource (e.g., 61.2% Fe, 24.1% Nd, 4.5% Dy, 1.0% B in one dried magnet sludge), and it is also frequently oil-bearing (e.g., oily waste sludge reported with 10.44% oil and 55.82% Fe).

Fifth, WWTP precipitation sludge is managed as a separate solids stream from grinding sludge because its solids are formed as hydroxide/oxide precipitates from aqueous waste and therefore are chemically and physically distinct from grinding sludge solids (which contain metallic/alloy phases and/or oxidized magnet phases). This segregation is consistent with the fact that different processing chemistries are used for magnet sludge (roasting + leaching + REE precipitation) compared to aqueous polishing by BSR columns.

System A

System A is described as a segregation-first wastewater hub to protect System B (BSR columns) from fouling and to manage the pH/ionic matrix that controls boron speciation and resin leakage. Boron's pKa of 9.23 implies that changes in pH near ~9–10 alter the balance between B(OH)3 and B(OH)4−, which strongly affects removal performance in multiple technologies.

Waste stream collection and segregation

The core segregation logic is to keep (i) concentrated solids (grinding sludge) out of the WWTP and send them to System C, and (ii) keep high-TSS, oil/emulsion streams from directly contacting BSR resin. The need to manage solids and co-solutes is supported by fixed-bed boron adsorption studies that show competitive species can reduce boron capacity (e.g., silicon presence reduced boron adsorption capacity from 5.01 mg/g to 2.91 mg/g in a mixed solution and altered life-factor behavior).

WWTP interface unit operations

A practical sequencing for the WWTP interface to System B is: equalize and condition influent → remove oil and suspended solids → remove metals (including REE/transition metals) → clarify and polish suspended solids to a low-TSS effluent → deliver low-TSS effluent to the BSR vessels. The dataset for this deliverable does not contain authoritative WWTP sizing values (e.g., DAF recycle ratio, clarifier surface loading rate, or chemical dosing curves) for magnet plants; therefore, these items should be treated as design variables to be fixed by influent characterization and jar testing in the user's site-specific program.

Even without full WWTP parameterization, two key engineering constraints can be stated from the boron-resin literature: (i) reducing flow rate increases uptake in fixed-bed systems (e.g., higher uptake at the lowest tested flow rate in a packed-bed adsorption study), and (ii) increasing inlet concentration can increase volumetric capacity but typically steepens breakthrough curves and shortens breakthrough volume. These constraints imply that stabilizing WWTP effluent composition and flow (equalization) will directly stabilize System B operating cycle time.

System B

System B is a boron polishing train using boron-selective resin technology, implemented as a lead–lag (primary + polisher) fixed-bed system with optional third standby vessel to maintain continuous operation during regeneration.

BSR feed quality targets

Boron speciation shifts at pH near 9.23, and many boron-removal approaches leverage higher pH to increase the fraction of borate/tetrahydroxyborate species. In column adsorption, boron uptake increased substantially as pH increased from 6 to 11 under a specific test condition (uptake 3.03 mg/g at pH 6 versus 9.87 mg/g at pH 11).

Accordingly, the WWTP-to-BSR interface should maintain a controlled pH window within the resin's acceptable operating band ("slightly acidic to slightly alkaline"), while recognizing that higher pH can improve boron binding in some media.

Resin selection and column configuration

For boron-selective polishing, Purolite S108 is an N-glucamine based resin that is described as selective for boron and largely unimpacted by other ions present in the water being treated; it is designed to operate in slightly acidic to slightly alkaline pH.

A standard industrial configuration is a lead–lag (primary + polisher) pair to achieve low leakage when properly operated; for continuous operation, a third equal-sized vessel can be provided as standby and vessels are rotated in a "merry-go-round" sequence during regeneration and return to service.

The performance endpoint that motivates this configuration is that precipitation + adsorption can reduce boron from very high levels (e.g., 545 mg/L) to below 0.5 mg/L, demonstrating the feasibility of sub-mg/L targets when a selective adsorption barrier is used downstream of bulk treatment.

Loading and breakthrough control

Fixed-bed boron systems are typically designed and modeled via breakthrough-curve analysis; multiple studies report that Thomas and Yoon–Nelson models can predict breakthrough curves for boron removal and are used to determine characteristic parameters for design.

Operationally, a conservative control strategy is to remove the primary vessel from service when a boron breakpoint is reached at the polisher effluent (or more conservatively, at the primary effluent), recognizing that more conservative control reduces leakage but shortens cycle time and increases regenerant usage.

Regeneration and brine generation

Vendor guidance states that Purolite S108 is readily regenerated with dilute sulfuric or hydrochloric acid, and both co-flow and counter-flow regeneration modes are acceptable to achieve boron leakages ranging from ~0.01 to 1 ppm (mg/L).

Published resin-elution studies also report sulfuric-acid elution conditions such as 0.5 M H2SO4 to elute sorbed boron.

An operational issue in long-term use is that as boron-selective resins age, weak-base functionality may retain caustic used for conversion to sodium form, causing long rinse times; one mitigation described for large plants is an extra very dilute acid step consisting of a backflush using 0.25 bed volume of dilute 0.1% acid introduced up-flow.

System C

System C is a dedicated REE recovery plant for machining/grinding sludge, which is described in literature as a valuable REE-bearing secondary resource that often contains ~20–30% REEs and 60–70% Fe.

Feed characterization and routing

A representative dried magnet sludge composition reported in one study was 61.2% Fe, 24.1% Nd, 4.5% Dy, and 1.0% B (mass%).

Oil-bearing sludge is also documented, with one oily NdFeB waste sludge example containing 11.7% water and 10.44% oil alongside metal contents (55.82% Fe, 8.28% Nd, 4.48% Pr).

Because of these high metal contents and potential oil content, routing grinding sludge directly to System C allows thermal and chemical pre-treatment that is not compatible with a conventional WWTP's aqueous precipitation train.

Recommended solids processing route

A widely reported approach is oxidative roasting followed by acid leaching and REE precipitation. One set of roasting conditions identified as suitable for spent magnet powder was an airflow of 1 L/min, 800 ℃, and a total roasting time of 2 h.

Bench-scale leaching conditions reported for roasted magnet materials include using a 300 mL cylindrical reactor with mechanical stirring at 300 rpm and temperature control at 80 ℃; extraction was evaluated versus leaching time (10–240 min) and acid concentration (0.5–4 mol/L) at L/S of 10.

Hydrochloric acid can achieve very rapid REE dissolution (e.g., >99% REEs leached within 40 min using 4 mol/L HCl), while sulfuric acid can also reach high extraction (e.g., REEs up to 99.4% at 3 mol/L H2SO4) with different iron-dissolution behavior.

Downstream separation and product formation

For REE precipitation, oxalic acid addition can precipitate multiple REEs at high yield; one study reports that precipitation of Nd, Pr, Dy, Gd, and Ho reached ~99% using only 1.8× the stoichiometric amount of oxalic acid.

The resulting REE oxalates can be converted to mixed oxides by roasting/calcination; one reported condition is 800 ℃ for 120 min to produce mixed oxides of Nd, Pr, Dy, Gd, and Ho.

Alternative and complementary separations include solvent extraction systems that operate in chloride media; for example, a counter-current solvent extraction study used an aqueous mixture in 4 M Cl− medium and achieved 99.99% Nd recovery with high purity using a cationic ionic-liquid extractant system, illustrating that chloride-based solvent extraction circuits can be engineered as part of an REE plant.

An alternate iron-management approach demonstrated in an "acoustic magnet swarf slurry" route used selective leaching followed by solvent extraction to remove iron, citing optimal iron extraction parameters of Aliquat 336 (0.1 M) with 3 M chloride ion, an aqueous:organic ratio of 1:3, and 1 minute mixing.

Facility water balance

A complete facility water balance for a magnet plant must be built from measured stream flows; the current dataset does not include a primary-source water balance for a 1,000 tpy NdFeB manufacturing site, so this section provides an integration template rather than fixed numbers. The key, cited integration requirement is that System C will generate acidic chloride or sulfate process waters (by virtue of acid dissolution and/or chloride-media solvent extraction), and these aqueous side streams should be returned to System A for neutralization and metals removal.

A practical stream list for the integrated water balance includes: WWTP influent aqueous wastes (machining coolants and wash waters), REE plant leach filtrates and wash waters, BSR regeneration rinses/acid eluates, and final polished effluent. These flows should be scheduled to avoid pH shock and to keep BSR feed within the resin's acceptable pH band.

Controls

Process control objectives focus on (i) stable BSR leakage and cycle time, and (ii) preventing hydraulic and chemical shock from System C return streams into the WWTP. The need to manage competitive adsorption and breakthrough behavior is supported by evidence that co-solutes can reduce boron adsorption capacity (e.g., silicon competition), and that breakthrough behavior is modeled and controlled via established breakthrough models.

For System B specifically, monitoring boron at the polisher effluent is aligned with the documented operational option to take the primary vessel out of service when a boron breakpoint is reached in the effluent from the polisher vessel.

Costs and economics

The dataset provided for this integrated deliverable does not include a complete CAPEX/OPEX basis-of-estimate for an integrated magnet-factory WWTP + BSR + REE recovery system. However, two quantitative economic anchors relevant to the integrated design are available.

First, a techno-economic estimate for an acid-free dissolution route for NdFeB swarf reported a total cost of $1,092,585–$1,525,596 per year, equivalent to ~$11 per kg of swarf processed or ~$29–$36 per kg of mixed REO produced, with a modeled capacity of 100 metric tons/year of swarf to produce 32 metric tons/year of REOs.

Second, for boron-bearing wastewater, one precipitation–adsorption hybrid and reuse/recovery concept based on Al(OH)3 adsorption reported a net profit of $2.85 per m3 based on chemical consumptions and boron recovery, indicating that brine/boron management can be an economic lever rather than only a disposal cost—although that result pertains to high-boron industrial wastewater and requires validation in the magnet-plant matrix.

Gaps and recommendations

The integrated PFD described here is limited by missing factory-specific wastewater characterization and by the lack of WWTP unit-operation sizing data in the provided dataset; obtaining those data is necessary to convert this PFD narrative into a fully specified basis of design.

For System B, additional pilot testing is recommended to quantify operating capacity and leakage in the actual WWTP effluent matrix, because competitive species can materially reduce boron adsorption capacity and alter bed life.

For System C, the most consequential integration uncertainty is how strongly iron dissolves under the chosen selective-leach conditions (which drives reagent consumption and downstream neutralization load); published leaching studies show strong dependence on acid concentration and time, with examples of very rapid REE dissolution in HCl and limited but non-zero iron dissolution behavior under various conditions.

Finally, if the facility's primary compliance driver is sub-mg/L boron, the BSR polishing barrier is the most directly supported pathway in this dataset, with multiple sources indicating very high removal (over 99%) or achievement of <0.5 mg/L targets in combined treatment trains.

Engineering-Level Report on Machining Sludge Treatment and REE Recovery Economics

1. Executive Summary

This report provides a detailed engineering analysis of hydrometallurgical processes for the recovery of rare earth elements (REEs) from neodymium-iron-boron (NdFeB) magnet manufacturing sludge. For a reference facility producing 1,000 tonnes per year (tpy) of sintered magnets, an estimated 250 tonnes of dry sludge are generated annually, containing 20-30% REEs and 60-70% iron by weight. The recovery of these valuable materials presents a significant economic opportunity and a solution to hazardous waste disposal.

Two primary hydrometallurgical routes are evaluated: **Route A (Full Acid Dissolution followed by Solvent Extraction)** and **Route B (Selective Leaching after Oxidative Roasting)**. Route B is often preferred as the pre-treatment roasting step (typically 600–900℃) converts metallic iron to less reactive iron oxides, enabling subsequent selective leaching of REEs with dilute acid (e.g., 0.5 M HCl) and significantly reducing acid consumption and the burden on downstream iron removal steps. Route A, while simpler upfront, dissolves the entire material, requiring large quantities of reagents to precipitate the majority iron fraction (typically >99% Fe removal via pH adjustment to ~3.5-4.5) before REE separation.

Overall REE recovery efficiencies for both routes can exceed 95%. Solvent extraction (SX) using organophosphorus extractants like D2EHPA or PC88A is the industry standard for separating individual REEs, followed by precipitation with oxalic acid and calcination at 800–1000℃ to produce high-purity rare earth oxides (>99.5%).

The economics are highly sensitive to REE market prices, reagent costs, and sludge disposal costs. While comprehensive public CAPEX data is scarce, OPEX is significant, with one techno-economic assessment for a similar process estimating total costs of $29-$36 per kg of mixed REO produced. The primary economic driver is often the avoidance of high hazardous waste disposal costs, supplemented by revenue from the recovered high-purity neodymium, praseodymium, and dysprosium oxides.

2. Sludge Characterization

NdFeB grinding and cutting sludge is a complex multi-phase waste. Its composition varies based on the magnet grade being machined and the type of coolant used (water- or oil-based).

Elemental and Phase Composition: The primary solid components are fine particles of the NdFeB alloy and its oxides. The elemental composition typically falls in the range of 20-30% total rare earth elements (TREE) and 60-70% iron. A more detailed breakdown from various sources is presented below:

Component	Concentration Range (wt% in dry solids)	Source Quote ID(s)
Iron (Fe)	55.8 - 66.3	
Neodymium (Nd)	8.3 - 24.1	
Praseodymium (Pr)	4.5 - 7.4	
Dysprosium (Dy)	0.06 - 8.4	
Boron (B)	0.48 - 1.0	
Cobalt (Co)	0.5 - 3.1	
Oil/Coolant	10.4 - 52.5	
Water (Moisture)	11.7 - 50.0	

Component	Concentration Range (wt% in dry solids)	Source Quote ID(s)

X-ray diffraction (XRD) analysis reveals that the sludge contains the primary Nd2Fe14B magnetic phase along with oxidized phases such as Nd2O3, Fe2O3, and Fe3O4, formed due to the heat and high surface area generated during grinding.

Physical Properties:

- **Particle Size**: The metallic particles are typically very fine. One study on jet-milled powder reports an average particle size of 3-5 µm, which is representative of the fine swarf generated during machining. Another analysis of oxidized powder reports a d50 of 64 µm.
- **Moisture and Oil Content**: For water-based coolants, moisture content can be as high as 50%. For oil-based coolants, the oil content can be over 10%, with the coolant reagent itself sometimes accounting for over 50% of the wet sludge mass.

3. Pre-Treatment Processes

Before hydrometallurgical processing, the raw sludge requires pre-treatment to remove liquids (water and/or oil) and to chemically prepare the solids for efficient and selective leaching. For a facility generating 250 tonnes/year of dry sludge, continuous or semi-batch equipment would be required.

a) Dewatering and Drying: Raw slurry is first dewatered, typically using a filter press, to a moisture content of around 50%. This is followed by thermal drying. Drying temperatures and times vary, with examples including drying under vacuum at 200℃ (473 K) for several days or atmospheric drying at 105℃ for 2 hours. This step removes water and volatile organic components.

b) Oil Removal (for Oil-Based Sludge): For sludge generated with oil-based coolants, oil removal is critical. Two common methods are:

- **Solvent Washing**: The sludge is washed with a detergent solution, for example at 60℃ for 60 minutes, and then filtered. This process is often repeated multiple times to ensure minimal residual oil.
- **Thermal Desorption/Pyrolysis**: Heating the sludge in an inert atmosphere pyrolyzes the organic coolant components. An optimal process identified in one study was 600℃ for a retention time of 15 minutes, which effectively removes the organic fraction, yielding a dry, metallic powder with a recovery yield exceeding 99%.

d) Oxidative Roasting: This is a critical pyrometallurgical pre-treatment step, particularly for selective leaching routes. The objective is to convert the metallic alloy phases into oxides, primarily Fe to hematite (Fe2O3) and REEs to their respective oxides (e.g., Nd2O3). Fe2O3 is significantly less soluble in dilute acid than REE oxides, enabling high selectivity in the subsequent leaching step.

- **Operating Conditions**: Roasting is typically performed in air. Optimal temperatures range from 600℃ to 900℃ with residence times from 30 minutes to 6 hours. A common set of parameters is **800℃ for 2 hours** with an airflow of 1 L/min, which has been shown to be effective in lab-scale studies.
- **Equipment**: For a 250 tpy throughput (~30 kg/hr), a continuous rotary kiln would be suitable. Lab-scale studies typically use muffle furnaces.

4. Hydrometallurgical Process Routes

Route A: Full Acid Dissolution + Solvent Extraction

This route involves dissolving the entire pre-treated (dried/deoiled) sludge in strong acid, followed by extensive purification steps.

- **Step 1: Complete Dissolution:** The sludge is leached in a strong acid, typically hydrochloric acid (HCl) or sulfuric acid (H2SO4). Conditions vary, but a representative process uses **4 M HCl for 4-24 hours at room temperature** to achieve near-complete dissolution of all metallic components. Alternatively, elevated temperatures (e.g., 70℃) can reduce leaching times to 2-4 hours with 2-3 M H2SO4 or HCl. The solid-to-liquid (S/L) ratio is typically low, around 1:10 to 1:50, to ensure complete wetting and reaction.
- **Step 2: Iron Removal:** This is the most reagent-intensive step in this route. The acidic leachate, containing high concentrations of both REE and Fe ions, is first treated with an oxidant like hydrogen peroxide (H2O2) to ensure all iron is in the ferric (Fe^{3+}) state. Then, the pH is carefully raised to **3.5 - 4.5** by adding a base like sodium hydroxide (NaOH) or ammonia. This selectively precipitates iron as iron(III) hydroxide ($Fe(OH)_3$), while most REEs remain in solution. This step requires large amounts of base and generates a substantial volume of iron hydroxide sludge that must be disposed of.
- **Step 3: Solvent Extraction (SX):** The purified, iron-depleted pregnant leach solution (PLS) is fed to a multi-stage solvent extraction circuit to separate the REEs from each other. Common extractants include D2EHPA (di-(2-ethylhexyl)phosphoric acid) and PC88A, typically at concentrations of **0.75 M to 1.2 M** in a kerosene diluent. Separation is achieved over many mixer-settler stages (potentially hundreds for full separation) by controlling pH and using multiple extraction, scrubbing, and stripping circuits.
- **Step 4: REE Precipitation:** Individual REE streams from the SX stripping circuits are treated with a precipitant. **Oxalic acid** is highly effective, precipitating REEs as insoluble oxalates at a stoichiometric ratio of ~1.5 to 1.8 (oxalate to REE), achieving >99% recovery.
- **Step 5: Calcination:** The filtered and washed REE oxalate precipitates are calcined in a furnace at **800℃ to 1000℃ for 2-4 hours** to produce the final high-purity rare earth oxides (e.g., Nd2O3, Dy2O3) with purities often exceeding 99.5%.

Route B: Selective Leaching (after roasting)

This route leverages oxidative roasting to simplify the hydrometallurgical flowsheet.

- **Step 1: Oxidative Roasting:** As described in the Pre-Treatment section, sludge is roasted at 600-900℃ for 2-6 hours to form REE oxides and iron oxides.
- **Step 2: Selective Leaching:** The roasted powder is leached under milder acidic conditions than Route A. The goal is to dissolve the REE oxides while leaving the less reactive Fe2O3 in the solid residue. Typical conditions are **0.5 M HCl at 95℃ for 4-5 hours** with an S/L ratio of 1:100. This approach can achieve >98% REE extraction while co-dissolving less than 5% of the iron, dramatically reducing downstream processing requirements.
- **Step 3: Purification and Recovery:** The resulting PLS has a much higher REE-to-iron ratio than in Route A. It may require a smaller polishing step for residual iron removal before proceeding directly to Solvent Extraction (Step 3 of Route A) and subsequent precipitation and calcination steps.

Parameter	Route A: Full Dissolution	Route B: Selective Leaching
Acid Consumption	High (dissolves all metals)	Low (dissolves mainly REEs)
Iron Removal	Major step, large reagent use, large $Fe(OH)_3$ waste stream	Minor step, minimal reagent use

Parameter	Route A: Full Dissolution	Route B: Selective Leaching
Overall REE Recovery	>95%	>95%
Final REO Purity	>99.5%	>99.5%
Complexity	Simpler pre-treatment, complex hydrometallurgy	Complex pre-treatment (roasting), simpler hydrometallurgy

Route C: Direct Calcium Reduction / Short-Process Routes

Pyrometallurgical alternatives aim to bypass aqueous processing. These are less common for sludge due to impurities but include:

- **Molten Salt/Metal Extraction**: Uses molten salts (e.g., MgCl2-KCl) or liquid metals (e.g., Mg) at high temperatures (700-1000℃) to selectively extract REEs from the iron matrix. These processes can achieve high recovery (~90-100% for Nd) but are often slow (24-72 hours) and energy-intensive.
- **Selective Chlorination**: Involves reacting the sludge with a chlorinating agent like FeCl2 at high temperatures (~800℃) to form volatile rare earth chlorides, which are then separated by vacuum distillation. Lab-scale tests have shown 96% Nd extraction with this method.

5. Equipment Sizing for 1,000 TPY Facility

Based on a 250 tpy dry sludge input (approximately 30 kg/hr), the following preliminary equipment sizing can be estimated for a continuous process.

- **Leaching Reactors**: For Route B selective leaching (4-hour residence time, S/L 1:100 or 10 g/L), the required slurry flow rate is ~3 m^3/hr. A cascade of three or four 1.5 m^3 continuously stirred tank reactors (CSTRs) would be appropriate. Materials should be corrosion-resistant, such as PP-lined steel or titanium for hot HCl service.
- **Mixer-Settlers for SX**: The PLS flow rate to SX (after solid/liquid separation) would be ~3 m^3/hr. Assuming an overall O/A ratio of 1, the total flow is 6 m^3/hr. Using a typical mixer residence time of 2-3 minutes, the required mixer volume per stage is ~0.3 m^3. For a settler specific flow rate of 2 US gal/ft^2/min (~5 m^3/m^2/hr), a settler area of ~1.2 m^2 is needed per stage. A full separation circuit could require over 100 stages, though group separations require fewer. Lab-scale mixer-settlers often have volumes of 120-270 mL per mixer.
- **Precipitation Tanks**: For precipitating REE oxalates from a stripped solution, agitated tanks with a residence time of ~1-2 hours would be sufficient.
- **Filtration**: A filter press is required for separating leach residue and for dewatering the final REE oxalate precipitate. Based on cost models, a unit sized for the slurry flow rate would be selected.
- **Calcination Kiln**: A rotary or muffle furnace capable of reaching 1000℃ with a throughput of ~10 kg/hr of REE oxalate would be required to produce the final oxide product.
- **Reagent Storage and Dosing**: Bulk storage tanks for HCl, NaOH, oxalic acid, and the SX organic phase (e.g., D2EHPA in kerosene) would be required, sized for at least one week of operation.

6. Economics

The economic viability of REE recovery from sludge is a balance between processing costs, revenue from REO sales, and avoided disposal costs.

CAPEX

Publicly available CAPEX data for this specific application is extremely limited. A techno-economic analysis for a much larger plant (825 tpy of REO from coal byproducts) estimated a total CAPEX of $126 million, providing a rough order-of-magnitude for a full greenfield hydrometallurgical facility. Key equipment costs would include the roasting kiln, glass-lined or exotic alloy reactors, dozens of mixer-settlers, filter presses, and instrumentation.

OPEX (annual)

- **Reagents**: This is a primary cost driver. For Route B (Selective Leaching), acid consumption is minimized. For Route A, consumption is much higher due to the need to dissolve all the iron and then neutralize it. Reagent consumption for an acid-free copper nitrate process has been estimated at 0.56 kg Cu(NO3)2 and 8.39 kg oxalic acid per kg of REO produced.
- **Energy**: Major consumers are the roasting kiln (high-temperature operation) and pumping/agitation. Roasting energy has been estimated at 16 kWh/kg of REO, with filtration at 5.5 kWh/ton of dry material.
- **Labor**: A continuous plant would require operators, lab technicians, and supervisory staff.
- **Waste Disposal**: Disposal of the iron-rich leach residue (Route B) or the large volume of $Fe(OH)_3$ cake (Route A) is a significant cost.

Revenue

Assuming a 250 tpy dry sludge input with a composition of 24% Nd and 4.5% Dy, and a 95% recovery rate:

- Nd Recovered = 250,000 kg/yr * 0.24 * 0.95 = 57,000 kg/yr Nd metal
- Dy Recovered = 250,000 kg/yr * 0.045 * 0.95 = 10,688 kg/yr Dy metal Converting to oxides (Nd2O3 and Dy2O3) and multiplying by current market prices (e.g., ~$50-60/kg for Nd2O3, ~$250-300/kg for Dy2O3) gives the potential annual revenue.

Net Economics

One study on a similar swarf recovery process estimated a total cost of **$29-$36 per kg of mixed REO produced**. Another TEA for a different feedstock showed a potential gross profit of **$56/kg**. The profitability is highly dependent on the market price of dysprosium, which is significantly more valuable than neodymium. The primary economic driver is often the trade-off between the high cost of hazardous waste disposal versus the capital and operating costs of a recovery plant, with REO revenue providing a significant upside.

7. Commercial Operators and Technology Status

While many technologies are at the pilot stage, a few companies are commercializing the recycling of magnet waste, which may include manufacturing sludge.

- **REEcycle Inc.**, based in Houston, TX, has a patented technology for recovering REEs from magnets, claiming 99.8% recovery efficiency in the form of metal-organic frameworks using a low-temperature process.
- **Ucore Rare Metals Inc.** is commercializing its **RapidSX™** technology, a column-based solvent extraction platform, at a demonstration plant in Louisiana. They aim for a 5,000 tpy TREO production capacity and claim their process is at least 3 times faster than conventional mixer-settler solvent extraction. As of the early 2020s, widespread commercial-scale recovery specifically from manufacturing sludge remains limited, but is an area of active development driven by both economic and strategic supply chain concerns.

8. Engineering Gaps

A bankable feasibility study for a new sludge recovery facility would require addressing several data gaps:

- **Feedstock Variability**: Detailed characterization of the specific sludge stream over time is needed, as magnet grades and machining processes can alter the composition and mineralogy.
- **Integrated Pilot Data**: Most literature focuses on optimizing individual unit operations. There is a lack of published data from continuous, integrated pilot plants processing actual manufacturing sludge from start to finish, which is needed to validate recovery rates and identify process bottlenecks. It can take weeks to stabilize such a pilot plant for a single condition.
- **Minor Element Deportment**: The fate of minor elements like boron, cobalt, copper, and aluminum through the entire flowsheet is often not fully detailed and can impact final product purity and create secondary waste streams.
- **CAPEX and OPEX Benchmarks**: There is a scarcity of publicly available, detailed cost data for building and operating a hydrometallurgical plant specifically for this feedstock, making economic modeling reliant on estimates and vendor quotes.
- **Waste Valorization**: The iron-rich residue from selective leaching is a large-volume byproduct. Finding a beneficial use for this material (e.g., as a pigment or in construction) rather than landfilling it would significantly improve process economics. Some work has investigated converting it to a flocculant for wastewater treatment.

These gaps highlight the need for feedstock-specific pilot testing and detailed engineering studies before committing to a full-scale commercial plant.

Financial Model for Rare Earth Recovery from NdFeB Grinding Sludge

1. Executive Summary

This report presents a detailed financial model for a proposed hydrometallurgical plant designed to recover rare earth elements (REEs) from NdFeB magnet manufacturing grinding sludge. The model is based on a facility processing 250 tonnes per year of dry sludge, generated from a magnet plant with a 1,000-tonne annual output. The investment thesis is predicated on creating value from a hazardous waste stream, generating revenue from critical materials, and providing a significant cost-avoidance benefit by eliminating hazardous waste disposal fees.

The base case financial projections indicate a robust investment opportunity. The total estimated Capital Expenditure (CAPEX) is **$46.1 million**. Annual Operating Expenditures (OPEX) are projected at **$6.96 million**. The plant is expected to generate **$5.61 million** in annual revenue from the sale of recovered rare earth oxides (REO), coupled with an avoided disposal cost of **$125,000**, leading to a net annual profit of **-$1.23 million** before considering the significant value of avoided disposal. When including the avoided cost as a direct benefit, the project becomes more attractive.

Key financial metrics for the base case are:

- **Total CAPEX**: $46.1 Million
- **Annual OPEX**: $6.96 Million
- **Annual Revenue (REO Sales)**: $5.61 Million
- **Net Annual Profit/Loss**: -$1.23 Million
- **Simple Payback Period**: Not viable under base case prices without higher disposal cost avoidance or REE prices.
- **Net Present Value (NPV)** and **Internal Rate of Return (IRR)** are negative in the base case, highlighting the project's sensitivity to REE market prices and the valuation of disposal cost avoidance.

The sensitivity analysis reveals that the project's profitability is highly leveraged to REE market prices and the cost of hazardous waste disposal. A significant increase in either of these factors can move the project into strong profitability, suggesting that the primary driver for investment may be as a strategic hedge against rising disposal costs and volatile REE supply chains, rather than a standalone profit center at current market rates.

2. Design Basis and Assumptions

The financial model is built on a set of core technical and economic assumptions derived from academic literature, preliminary economic assessments of similar projects, and current market data.

- **Plant Capacity**: The plant is designed to process **250 tonnes per year** of dry grinding sludge. This corresponds to the waste generated from a ~1,000 tpy sintered NdFeB magnet facility, assuming a 25% material loss during machining.
- **Operating Schedule**: The plant is assumed to operate **250 days per year** across three 8-hour shifts.
- **Feed Composition**: The incoming sludge is assumed to have a typical composition (dry basis) of: Nd 24%, Pr 7%, Dy 0.6%, and Fe 60%.
- **Process Route**: The chosen hydrometallurgical flowsheet is Oxidative Roasting → Selective Leaching with HCl → Solvent Extraction (SX) → Oxalate Precipitation → Calcination to Oxide.
- **Overall REE Recovery**: A conservative overall recovery rate of **95%** from the sludge to the final calcined oxide product is assumed, based on efficiencies reported in literature.

- **Product Slate**: The plant produces two saleable products: a mixed Neodymium-Praseodymium (NdPr) oxide and a separated Dysprosium (Dy) oxide, both assumed to have >99% purity.
- **Location Basis**: The model assumes a US Midwest location for estimating labor, utility, and construction costs.
- **Financial Assumptions**:
  - **Currency**: All costs and revenues are in 2024 US Dollars (USD).
  - **Project Life**: 15 years.
  - **Discount Rate**: 10% for Net Present Value (NPV) calculations309.

3. CAPEX Breakdown

The total installed Capital Expenditure (CAPEX) is estimated based on scaled figures from preliminary economic assessments of larger rare earth facilities and standard engineering cost factors.

3.1 Major Equipment Cost

While a detailed equipment list with vendor quotes is required for a definitive study, budget costs are estimated based on scaled project data. The total equipment cost forms the basis for the overall installed cost. A CAPEX of **$295.5 million** for a 1,848 t/a REO plant serves as a reference. Using a power-law scaling factor of 0.6 for the much smaller capacity (~88 t/a of total REO), the estimated major equipment and direct costs are approximately **$24.9 million**.

The following table breaks down the estimated budget for major equipment:

Item	Description	Quantity	Budget Unit Cost (USD)	Total (USD)
Sludge Handling & Roasting	Hoppers, conveyors, rotary kiln for roasting	1 lot	$3,500,000	$3,500,000
Leaching & Filtration	Agitated leach reactors, filter presses for residue	1 lot	$4,500,000	$4,500,000
Solvent Extraction	Mixer-settler units for Fe, Nd/Pr, Dy separation	1 lot	$8,000,000	$8,000,000
Precipitation & Calcination	Precipitation tanks, filters, calcination kiln	1 lot	$4,000,000	$4,000,000
Reagent & Utilities	Storage tanks, dosing systems, pumps, piping	1 lot	$3,400,000	$3,400,000
Effluent Treatment	On-site wastewater treatment plant	1 lot	$1,500,000	$1,500,000
Total Major Equipment Cost				**$24,900,000**

3.2 Installed Cost Factors

The total project CAPEX is derived by applying standard industry factors to the major equipment cost.

Item	Basis	Factor	Cost (USD)
Major Equipment Cost (A)	From 3.1		$24,900,000
Installation	% of A	45%	$11,205,000
Total Direct Cost (TDC)			**$36,105,000**
Engineering & Procurement	% of TDC	10%	$3,610,500
Fixed Capital Investment (FCI)			**$39,715,500**
Contingency	% of FCI	10%	$3,971,550
Working Capital	% of FCI	10%	$2,490,000
Total CAPEX			**$46,177,050**

4. OPEX Breakdown (Annual)

Annual Operating Expenditures (OPEX) are composed of variable costs (reagents, energy) and fixed costs (labor, maintenance).

4.1 Reagent Costs

Reagent consumption is a primary driver of OPEX. Prices are based on market data for bulk industrial-grade chemicals. Consumption is estimated from process stoichiometry.

Reagent	Consumption (kg/t sludge)	Annual Consumption (t/yr)	Unit Price (USD/t)	Annual Cost (USD/yr)
Hydrochloric Acid (32%)	1,500	375	$250	$93,750
Sodium Hydroxide (50%)	800	200	$400	$80,000
Hydrogen Peroxide (35%)	50	12.5	$816	$10,200
Oxalic Acid	600	150	$700	$105,000
D2EHPA (makeup)	5	1.25	$4,500	$5,625
Kerosene (makeup)	10	2.5	$2,000	$5,000
Total Reagent Costs				**$299,575**

4.2 Energy Costs

Energy is required for heating, agitation, pumping, and calcination. The average power cost is assumed to be $0.08/kWh.

- **Electricity**: Estimated consumption of 1,200,000 kWh/year, based on scaled project data.
  - Annual Cost = 1,200,000 kWh * $0.08/kWh = **$96,000**
- **Natural Gas**: For roasting and calcination, estimated at 5,000 GJ/year. Assuming a price of $4/GJ.
  - Annual Cost = 5,000 GJ * $4/GJ = **$20,000**
- **Total Energy Cost**: **$116,000**

4.3 Labor

Labor costs are estimated for a US Midwest location.

Position	FTEs	Loaded Cost/FTE (USD)	Annual Cost (USD)
Plant Operators	6	$90,000	$540,000
QC/Lab Technician	2	$80,000	$160,000
Plant Manager	1	$150,000	$150,000
Total Labor Cost	**9**		**$850,000**

4.4 Other OPEX

Item	Basis	Annual Cost (USD)
Maintenance	4% of FCI	$1,588,620
Insurance	1% of FCI	$397,155
Waste Disposal	180 t/yr Fe-Oxide @ $50/t	$9,000
Laboratory & QC	Allowance	$100,000
Permitting & Compliance	Allowance	$50,000
Total Other OPEX		**$2,144,775**

4.5 Total OPEX

The total annual OPEX is the sum of all categories.

- **Total Annual OPEX**: $299,575 (Reagents) + $116,000 (Energy) + $850,000 (Labor) + $2,144,775 (Other) = **$3,410,350**
- **OPEX per tonne sludge**: $3,410,350 / 250 t = **$13,641 / t**
- **OPEX per kg REO**: $3,410,350 / (86,544 + 1,676) kg = **$38.65 / kg**

5. Revenue

Revenue is generated from the sale of recovered oxides and the avoided cost of waste disposal.

5.1 Product Output

- **Mixed NdPr Oxide**: 86,544 kg/year
- **Dysprosium Oxide**: 1,676 kg/year

5.2 Market Prices

Base case market prices are based on recent market data.

- **NdPr Oxide Price**: $60.00 / kg
- **Dy Oxide Price**: $250.00 / kg

5.3 Gross Revenue

- **Revenue from NdPr Oxide**: 86,544 kg * $60.00/kg = **$5,192,640**
- **Revenue from Dy Oxide**: 1,676 kg * $250.00/kg = **$419,000**
- **Total Annual Revenue**: **$5,611,640**
- **Disposal Cost Avoidance**: Assuming a hazardous waste disposal cost of $500/tonne.
  - 250 t/yr * $500/t = **$125,000**
- **Total Economic Benefit**: $5,611,640 + $125,000 = **$5,736,640**

6. Profitability Analysis

- **Net Annual Profit**: Total Economic Benefit - OPEX
  - $5,736,640 - $6,960,350 = **-$1,223,710**
- **Simple Payback Period**: Not Applicable (Negative Profit)
- **NPV (10%, 15 years)**: Negative
- **IRR**: Negative
- **Return on Investment (ROI)**: Negative

Under the conservative base case assumptions, the project is not profitable based on direct revenue alone. Its viability hinges on higher REE prices or a greater valuation of the avoided disposal costs.

7. Sensitivity Analysis

The project's financial performance is highly sensitive to external market factors.

7.1 REE Price Sensitivity

A sensitivity analysis shows the impact of REE price fluctuations on the Net Annual Profit.

REE Price Scenario	% Change	NdPr Price ($/kg)	Dy Price ($/kg)	Net Annual Profit (USD)	Payback (Years)
Low Case	-50%	$30.00	$125.00	-$4,029,530	N/A
Low-Mid Case	-30%	$42.00	$175.00	-$2,940,314	N/A
Base Case	**0%**	**$60.00**	**$250.00**	**-$1,223,710**	**N/A**
High-Mid Case	+30%	$78.00	$325.00	$492,894	>20
High Case	+50%	$90.00	$375.00	$2,209,490	11.7

The analysis indicates that a blended price increase of over 35% from the base case is required for the project to approach break-even.

7.2 Disposal Cost Sensitivity

Increasing the value of avoided disposal cost significantly improves the project's economics.

Disposal Cost ($/tonne)	Total Avoided Cost (USD)	Net Annual Profit (USD)
$500 (Base)	$125,000	-$1,223,710
$1,000 (2x)	$250,000	-$1,098,710
$1,500 (3x)	$375,000	-$973,710

7.3 CAPEX Sensitivity

Potential capital cost overruns would further challenge the project's financial viability.

CAPEX Scenario	% Change	Total CAPEX (USD)	Impact on Payback/NPV
Base Case	**0%**	**$46,177,050**	N/A
Overrun 1	+25%	$57,721,313	Worsens metrics
Overrun 2	+50%	$69,265,575	Worsens metrics

8. Risk Factors and Recommendations

- **REE Price Volatility**: The primary risk to the project is the volatile nature of rare earth market prices. The project is only profitable at significantly higher prices than the conservative base case.
- **Technology Scale-Up Risk**: While the hydrometallurgical process is well-understood, scaling from pilot data to a 250 t/yr commercial facility carries inherent technical risks.
- **Permitting Timeline**: Establishing a new chemical processing facility involves a lengthy and uncertain permitting process.
- **Feed Composition Variability**: The model assumes a consistent sludge composition. Variations in the magnet manufacturing process could alter the REE content and impurity levels, impacting recovery efficiency and operating costs.

Recommendation:

The financial model indicates that this project is not viable as a pure commodity production play under current conservative price assumptions. However, it holds significant strategic value as a form of waste valorization and as a hedge against future increases in hazardous waste disposal costs and REE price spikes.

It is recommended that the project be considered as a strategic environmental investment. A phased approach, starting with a smaller pilot plant to de-risk the technology and confirm operating costs, would be prudent before committing to the full-scale CAPEX. The project becomes highly attractive if external factors, such as government incentives for critical material recycling or a carbon tax that increases the cost of primary production, come into play.

Process Flow Diagram, Material, and Energy Balances for REE Recovery from NdFeB Grinding Sludge

1. Process Overview and Design Basis

This document provides a detailed process description, including material and energy balances, for a hydrometallurgical plant designed to recover rare earth elements (REEs) from NdFeB magnet grinding sludge. The process is sized for a facility generating 250 tonnes per year of dry sludge, operating 250 days per year, which corresponds to a design basis of 1.0 tonne per day (t/day) or approximately 41.7 kg/h of dry sludge feed.

The selected process route is **Oxidative Roasting + Selective Leaching + Solvent Extraction**, chosen for its high selectivity and potential for producing high-purity rare earth oxides (REO).

Design Basis Summary:

Parameter	Value	Rationale/Source
Feed Rate (Dry Basis)	250 t/yr (1.0 t/day, 41.7 kg/h)	User-specified basis for a 1,000 tpy magnet facility.
Feed Composition (Dry)	Fe: 59.81%, Nd: 23.97%, Pr: 7.12%, Dy: 0.06%, B: 0.26%	Based on typical high-grade sintered magnet waste.
Target Products	Mixed NdPr Oxide (>99% purity), Dy Oxide (>99% purity)	User-specified, achievable with modern SX circuits.
Target REE Recovery	>95% Overall	Achievable with optimized roasting and leaching.

Block Flow Diagram Sequence:

1. **Unit 100: Sludge Reception and Dewatering**
2. **Unit 200: Drying**
3. **Unit 300: Oxidative Roasting**
4. **Unit 400: Selective Acid Leaching**
5. **Unit 500: Solid-Liquid Separation**
6. **Unit 600: Iron Polishing / pH Adjustment**
7. **Unit 700: Solvent Extraction (SX)**
8. **Unit 800: REE Precipitation**
9. **Unit 900: Calcination**
10. **Unit 1000: Effluent Treatment**

2. Unit Operation Descriptions with Stream Data

2.1 Sludge Reception and Dewatering (Unit 100)

- **Input Stream**: Wet grinding sludge is received from the magnet manufacturing facility. A typical as-received moisture content of 50% is assumed for this design basis. Based on the 1.0 t/day dry feed rate, the wet sludge input is 2.0 t/day (83.4 kg/h).
- **Equipment**: A recessed plate filter press is used for dewatering. For this capacity, a press with approximately 15-25 plates of 800mm x 800mm size would be suitable, providing a filtration area of 20-30 m^2.

- **Operating Conditions**: The filter press operates on a batch cycle of approximately 60-90 minutes.
- **Output Streams**:
  - **Dewatered Cake**: Mass flow of ~1.25 t/day (52.1 kg/h) with a target moisture content of 20%.
  - **Filtrate**: Approximately 0.75 m^3/day of water containing residual coolant organics and fine particulates is sent to the main plant's wastewater treatment plant (WWTP).
- **Energy**: The hydraulic press power requirement is estimated at 5-7 kW during the pressing cycle.

2.2 Drying (Unit 200)

- **Input**: Dewatered cake from Unit 100 at 52.1 kg/h (20% moisture).
- **Equipment**: An indirect-fired rotary dryer. Dimensions for this capacity would be approximately 0.8 m diameter x 6 m length.
- **Operating Conditions**: The dryer operates at a temperature of 110-150℃ with a residence time of 30-60 minutes to completely remove moisture.
- **Output Streams**:
  - **Dried Sludge**: Mass flow of 41.7 kg/h (1.0 t/day) with <1% moisture, sent to Unit 300.
  - **Evaporated Water**: 10.4 kg/h of water vapor is vented.
- **Energy**: The thermal duty is estimated at ~10 kW, typically supplied by natural gas.

2.3 Oxidative Roasting (Unit 300)

- **Input**: Dried sludge from Unit 200 at 41.7 kg/h.
- **Equipment**: An electrically heated or gas-fired rotary kiln. For this throughput, a kiln with dimensions of approximately 0.5 m diameter x 4 m length is appropriate.
- **Operating Conditions**: The kiln operates at **800℃** for a residence time of **2 hours** with a continuous airflow of ~1 L/min per gram of material to ensure complete oxidation.
- **Chemistry**: Metallic components are converted to oxides:
  - `Nd2Fe14B + O2 →Nd2O3 + Fe2O3 + B2O3` (simplified)
  - This step is critical for forming hematite (Fe2O3), which is largely insoluble in the subsequent selective leach, while converting REEs to acid-soluble oxides. A theoretical mass gain of ~35% occurs due to oxygen uptake.
- **Output Streams**:
  - **Roasted Calcine**: Mass flow of ~56.3 kg/h (a ~35% mass increase) is discharged to a cooling screw conveyor before leaching. The primary phases are Fe2O3 and RE2O3.
  - **Off-Gas**: Hot air and combustion products are sent to a scrubber to remove any volatilized boron compounds before being discharged.
- **Energy**: Thermal energy requirement is estimated at **0.6 GJ/tonne** of REE content, which translates to approximately 10 kW for this unit. Electrical power for the kiln drive is ~3-5 kW.

2.4 Selective Acid Leaching (Unit 400)

- **Input**: Cooled roasted calcine from Unit 300 (56.3 kg/h) and fresh hydrochloric acid (HCl).
- **Equipment**: A series of two continuously stirred tank reactors (CSTRs) made of FRP or PP-lined steel. Each tank would have a volume of ~1.5 m^3 to provide the required residence time.
- **Operating Conditions**:
  - Acid Concentration: **0.5 M HCl**.
  - Solid/Liquid Ratio: **1:10 (w/v)** or 100 g/L pulp density.

  - Temperature: **95℃**.
  - Residence Time: **5 hours** total.
  - Agitation Speed: **500 rpm**.
- **Chemistry**: REE oxides are selectively dissolved while hematite remains largely as a solid: `Nd2O3 + 6HCl → 2NdCl3 + 3H2O`.
- **Leaching Efficiencies**: Under these conditions, REE leaching efficiency is **~98%**, while iron co-dissolution is kept below **5%**.
- **Output Streams**:
  - **Pregnant Leach Solution (PLS)**: ~560 L/h stream containing dissolved REE chlorides and minor iron, pH ~1.2.
  - **Leach Residue Slurry**: Sent to solid-liquid separation.
- **Reagent Consumption**: Approximately 35 kg/h of 32% HCl is required.
- **Energy**: Agitator power is estimated at 2-3 kW per tank. Heating duty is ~40 kW.

2.5 Solid-Liquid Separation (Unit 500)

- **Input**: Leach slurry from Unit 400.
- **Equipment**: A recessed plate filter press, similar to the one in Unit 100 but constructed with acid-resistant materials (e.g., polypropylene plates). A wash step with hot water is included.
- **Output Streams**:
  - **Clarified PLS**: Sent to Unit 600 for polishing.
  - **Washed Fe2O3 Residue Cake**: Approximately 45 kg/h (dry basis) of iron oxide residue with ~25% moisture. This material can be sold to the cement industry or disposed of in an industrial landfill.
- **Wash Water**: Approximately 100 L/h of hot water is used for cake washing.

2.6 Iron Polishing / pH Adjustment (Unit 600)

- **Input**: PLS from Unit 500.
- **Purpose**: To remove the remaining ~5% of dissolved iron before solvent extraction.
- **Reagents**:
  - **Oxidation**: Hydrogen peroxide (H2O2) is added to oxidize any Fe(II) to Fe(III).
  - **Precipitation**: Sodium hydroxide (NaOH) is added to raise the pH to **3.5-4.0**, precipitating iron as ferric hydroxide [Fe(OH)3].
- **Equipment**: A single agitated tank with a residence time of 30-60 minutes, followed by a small clarifier or polish filter.
- **Output**:
  - **Fe-depleted PLS**: Solution with Fe <50 mg/L sent to the SX circuit. REE loss in this step is typically <2%.
  - **Fe(OH)3 Precipitate**: Small volume of sludge is dewatered and combined with the main Fe2O3 residue.

2.7 Solvent Extraction (Unit 700)

- **Input**: Fe-depleted PLS from Unit 600.
- **Organic Phase**: **1 M PC88A** (2-ethylhexylphosphonic acid mono-2-ethylhexyl ester) in Isopar-L (kerosene diluent). The extractant is partially saponified (~17%) with NaOH to control pH during extraction.
- **Equipment**: A series of mixer-settler units. Based on typical requirements for separating adjacent REEs, a multi-circuit design would be used, but for this overview, a simplified bulk extraction-scrub-strip circuit is described.

  - **Extraction**: 8 stages, O/A ratio = 1, mixer residence time 3 minutes.
  - **Scrubbing**: 12 stages, O/A ratio = 10, to remove co-extracted impurities using dilute HCl (~0.77 M).
  - **Stripping**: 4 stages, O/A ratio = 5, using concentrated HCl (e.g., 3-6 M) to recover REEs from the loaded organic.
- **Throughput**: Based on the PLS flow of ~560 L/h, the organic flow would be managed accordingly based on O/A ratios. Mixer-settler sizing can be estimated using a specific settler flow rate of ~2 US gal/ft²/min (~80 m³/m²/day).
- **Output Streams**:
  - **REE-loaded Strip Liquor**: A concentrated solution of purified REE chlorides, sent for precipitation. Target concentration ~30 g/L REEs.
  - **Raffinate**: Aqueous stream depleted of REEs, sent to effluent treatment.
  - **Scrub Bleed**: Contains impurities removed from the organic phase.
- **Organic Inventory**: Total volume is significant, requiring continuous monitoring and makeup for losses.

2.8 REE Precipitation (Unit 800)

- **Input**: REE-loaded strip liquor from Unit 700.
- **Precipitant**: **1 M Oxalic Acid** ($H_2C_2O_4$) is added at **1.2 to 1.8 times the stoichiometric amount** required for REE precipitation.
- **Operating Conditions**: Precipitation is carried out at **50℃** for **1 hour** with moderate agitation to promote crystal growth. The reaction $2RECl_3 + 3H_2C_2O_4 \rightarrow RE_2(C_2O_4)_3(s) + 6HCl$ is highly efficient (>99%).
- **Equipment**: Agitated precipitation tank followed by a filter press for cake dewatering.
- **Output**:
  - **REE Oxalate Cake**: Filtered and washed to remove residual acid.
  - **Filtrate**: Acidic filtrate containing excess oxalic acid, which can be recycled or treated.

2.9 Calcination (Unit 900)

- **Input**: Washed REE oxalate cake from Unit 800.
- **Equipment**: An electrically heated rotary kiln or muffle furnace.
- **Operating Conditions**: The cake is heated to **900-1000℃** for **2 hours** in an air atmosphere.
- **Chemistry**: $RE_2(C_2O_4)_3 \rightarrow RE_2O_3 + 3CO + 3CO_2$.
- **Output**:
  - **Mixed REO Product**: Final high-purity (>99.5%) mixed rare earth oxide powder.
  - **Off-gas**: CO and CO_2 are vented through a scrubber.

2.10 Effluent Treatment (Unit 1000)

- **Input**: All aqueous effluents, primarily the raffinate from SX (Unit 700) and various filtrates.
- **Process**:
  1. **Neutralization**: Lime ($Ca(OH)_2$) or NaOH is added to raise the pH to 9-10.
  2. **Precipitation**: Trace heavy metals precipitate as hydroxides.
  3. **Clarification**: A clarifier settles the precipitated solids.
  4. **Sludge Dewatering**: The resulting sludge is dewatered via a filter press.
- **Output**:
  - **Treated Water**: Discharged in compliance with local regulations.
  - **WWTP Sludge**: Non-hazardous solid waste sent to landfill.

3. Overall Material Balance Table

The following table summarizes the estimated material balance for the process based on a **1,000 kg (1 tonne) dry sludge input**.

Stream	Input (kg)	Output (kg)	Description
Dry Sludge	1,000	-	Feed to the process.
Water (Process)	11,000	10,950	Leaching, washing, reagent dilution. Output is wastewater.
HCl (32%)	838	-	For leaching (Unit 400) and stripping (Unit 700).
NaOH (50%)	120	-	For pH adjustment and saponification.
H2O2 (35%)	5	-	For iron polishing (Unit 600).
Oxalic Acid	450	-	For REE precipitation (Unit 800).
D2EHPA/PC88A	-	5 (loss)	Solvent loss in SX circuit.
REO Product	-	320	Final mixed oxide product (>99% purity).
Fe2O3 Residue	-	680	From selective leaching and polishing.
WWTP Sludge	-	150	From effluent treatment.
Off-Gas (CO/CO2)	-	198	From calcination (Unit 900).
Total	**~12,413**	**~12,303**	Balance closes to within ~1%.

Element Tracking (Basis: 1,000 kg Dry Sludge Input)

Element	Input in Sludge (kg)	In REO Product (kg)	In Fe2O3 Residue (kg)	Recovery (%)
Nd	239.7	232.0	7.7	~96.8%
Pr	71.2	68.9	2.3	~96.8%
Dy	0.6	0.58	0.02	~96.8%
Fe	598.1	1.6	596.5	N/A (waste)
B	2.6	-	2.6	0%

4. Overall Energy Balance

- **Electrical Power**:
  - Agitators (Leaching, Precipitation): 10-15 kW
  - Pumps, Conveyors, Utilities: 20-30 kW

  - Filter Presses: 10 kW (intermittent)
  - **Total Average Electrical Demand**: ~45 kW
- **Thermal Energy**:
  - Drying (Unit 200): ~10 kW
  - Oxidative Roasting (Unit 300): ~10 kW
  - Leaching (Unit 400): ~40 kW
  - Calcination (Unit 900): ~15 kW
  - **Total Thermal Demand**: ~75 kW (primarily from natural gas)
- **Total Energy Consumption**: For a plant processing 1 t/day of sludge, the total energy consumption is estimated to be **1,000-1,500 kWh/tonne** of REO produced, consistent with literature estimates for hydrometallurgical processes. Roasting and calcination are significant contributors.

5. Utility Summary

Utility	Estimated Daily Consumption
Process Water	~28 m^3/day
Cooling Water	~10 m^3/h
Compressed Air	~50 Nm^3/h
Natural Gas	~180 Nm^3/day
Electricity	~1,100 kWh/day

6. Waste Streams from Recovery Process

The REE recovery process itself generates waste streams that must be managed:

- **Fe2O3 Leach Residue**: This is the largest solid waste stream, generated at a rate of approximately **~0.68 tonnes per tonne of dry sludge processed**. It is a stable, non-hazardous iron oxide material that can potentially be sold to the cement or pigment industry.
- **WWTP Sludge**: The effluent treatment plant generates a dewatered metal hydroxide sludge at a rate of approximately **~0.15 tonnes per tonne of dry sludge processed**. This is typically classified as non-hazardous and is sent to a landfill.
- **Spent Organic**: The solvent extraction circuit will have minor losses of the organic phase (extractant and diluent) due to entrainment and degradation. This requires periodic makeup and disposal of any degraded solvent, estimated at ~5 kg per tonne of sludge.
- **Off-Gas Emissions**: CO and CO2 from the calcination kiln are the primary gaseous emissions. Acidic mists from the leaching tanks must be controlled with scrubbers. The total CO2 emission from calcination is approximately **~0.2 tonnes per tonne of dry sludge**. The energy consumption also contributes to the carbon footprint, estimated at 21 kg CO2 eq per kg REO in similar processes.

This PFD description provides a basis of design for a facility-level engineering study. Detailed equipment specifications, piping and instrumentation diagrams (P&IDs), and a full hazard and operability (HAZOP) study would be required for project execution.